Local–Global Spectral–Spatial Dual Deep Subspace Clustering Network for Hyperspectral Images

Han Zhai, *Member, IEEE*, and Jun Yu, *Student Member, IEEE*

***Abstract*—Deep subspace clustering (DSC) has become increasingly popular for hyperspectral image (HSI) interpretation, achieving remarkable accomplishments. However, the existing methods overemphasize spatial pattern mining but ignore the importance of spectral information, and insufficiently explore the underlying mechanism of self-expression learning, which results in limited clustering accuracy. To address these issues, this article proposed an innovative local–global spectral–spatial dual DSC (LGDSC) network for HSIs, which makes full use of the spectral and spatial discriminative information and integrates the advantages of local and global expressions to boost clustering performance. First, an effective dual convolutional autoencoder (CAE) architecture was developed to fully mine complementary local-spectral and global-spatial features through a 1-D CAE and 2-D CAE, respectively, with a shared self-expression module embedded for pixel affinity learning. Then, a novel selective affinity aggregation (SAA) module was put forward to precisely fuse favorable affinities derived by differentiated features to generate a more indicative and clustering-friendly adjacency matrix. Ultimately, a valid joint loss function with biliteral constraints was designed to guide local-spectral and global-spatial feature learning in the training process. Extensive experimental results on four public datasets clearly illustrate the significant advantage of LGDSC over the other state-of-the-art (SOTA) alternatives and its promising prospects in practical applications.**

***Index Terms*—Deep subspace clustering (DSC), dual convolutional autoencoder (CAE) architecture, hyperspectral image (HSI), joint loss function with biliteral constraints, selective affinity aggregation (SAA) module.**

I. Introduction

HYPERSPECTRAL remote sensing is of great significance in numerous application fields owing to its advanced ability to capture the subtle spectral diagnostic differences between ground objects by hundreds of narrow wavelength bands [1], [2], [3], [4], [5]. Unlike traditional panchromatic, red-green-blue (RGB) and multispectral images, hyperspectral images (HSIs) record detailed reflective properties of various ground objects to support subcategory classification owing to their nearly continuous spectral profiles. This characteristic makes HSIs widely applicable in various fields, such as mineral mapping [6], precision agriculture [7], and coastal zone monitoring [8]. Clustering plays a crucial role in unsupervised interpretation of HSIs and has attracted increasing attention. It partitions pixels into different groups by mining similarities in the data itself without using any training samples. However, the complex structure, obvious nonlinearity, and varied spectra of HSIs pose great challenges for clustering.

To date, numerous clustering approaches have been proposed for HSIs from different perspectives. These can be grouped into centroid-based, density-based, probability-based, biology-inspired, and graph-based models [9]. Specifically, centroid-based methods perform clustering by iteratively optimizing the centroids and their associated pixel assignments to minimize within-cluster squared error, such as k-means [10] and fuzzy c means [11]. Density-based methods identify clusters as high-density regions in the feature space, like density peak clustering [12] and diffusion-based clustering [13], which are generally robust to cluster shape and noise. Probability-based methods assume that pixels originate from a mixture of specific distributions and separate them according to probability rules, such as the Gaussian mixture model (GMM) [14] and expectation maximization [15]. Biology-inspired methods simulate HSI data based on biological models and generate clusters through genetic computation, like the immune system [16] and deoxyribonucleic acid (DNA) network [17]. Graph-based methods model pixel similarity through an affinity matrix constructed by handcrafted features and partition pixels by graph cutting, such as sequential spectral clustering [18], anchor graph clustering [19], and bipartite graph clustering [20]. However, these traditional methods suffer from limited clustering performance, since the complex structure of HSIs frequently violates their basic assumptions.

To overcome these obstacles, extensive subspace clustering methods have been developed. These approaches assume pixels lie in a union of subspaces rather than a single global space and learn the underlying partition pattern by exploiting the self-expression property of data themselves [21], [22], [23]. These models adapt well to the complex structure and spectral variations of HSIs to capture the intrinsic pixel relationships through subspace modeling and self-expression learning, which greatly advance the development of clustering. Among them, sparse subspace clustering (SSC) and low-rank subspace clustering (LRSC) are the two most popular fundamental models, which promote sparse and low-rank self-expression of pixels to exploit local and global structures of HSIs, respectively [24], [25], [26], [27]. One of the most typical representatives is spectral–spatial sparse subspace clustering

Received 26 August 2025; accepted 8 September 2025. Date of publication 12 September 2025; date of current version 2 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42271386 and Grant 42001313 and in part by the National Science and Technology Basic Resource Investigation Project under Grant 2019FY202503. *(Corresponding author: Han Zhai.)*

The authors are with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, Hubei 430074, China (e-mail: zhaihan@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3609665

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

(S^4C), which is a pioneering work in this field. It achieved impressive clustering results by constructing effective spectral and spatial constraints to improve the self-expression accuracy [21], [22]. Based on this method, extensive modified models were developed to further enhance the clustering performance by learning more precise affinities, such as kernel-based [28], object-based [29], dictionary-based [30], multiview feature [31], and graph convolutional regularization (GCR)-based methods [32]. Moreover, faced with the unbearable complexity of these self-dictionary-based methods, various scalable models were put forward to further improve their applicability to large HSIs, such as sparse coding clustering [33] and sketched subspace clustering [34]. These methods obtained outstanding performance and showed considerable potential in practice. Although these subspace clustering methods have achieved notable results and greatly improved the latent value of HSIs, they remain shallow linear models with limited discriminability and adaptability, which leads to degraded performance and reduced attractiveness in real-world scenarios.

The rapid development of deep learning has introduced a new solution to HSI clustering, known as deep subspace clustering (DSC). This approach capitalizes on the advantage of subspace clustering and alleviates certain limitations by employing the strong feature extraction and nonlinear representation abilities of deep learning. DSC presents an incomparable superiority to traditional arts and has become a recent area of focus, with a wide range of advanced models developed [35], [36], [37]. For instance, by constructing 3-D convolutional autoencoders (CAEs) to mine abstract and robust spectral–spatial features for self-expression, a set of CAE-based DSC methods has been proposed for HSIs, such as Laplacian regularized CAE (L-CAE) [38] and spectral–spatial attention enhanced CAE (3D-ACAE) [39], obtaining noticeable improvements in clustering accuracy. Since most autoencoders merely learn the implicit geometric structure of original samples while overlooking distribution information, a distribution-preserving DSC model (DSCOT) was developed, which embeds geometric distance between samples to further improve the clustering performance [40]. In addition, by introducing GCR to capture nonEuclidean geometric structures, many GCR-DSC models were put forward, which learn robust feature expressions to deliver state-of-the-art (SOTA) performance. Representative examples include graph regularized residual subspace clustering network (GR_RSCNET) [41], hypergraph structure autoencoder (HyAE) [42], and dual smooth graph convolutional clustering (DSGCC) [43]. Furthermore, to adapt to large HSIs, many contrastive learning (CL)-based DSC models were developed, such as two-branch CL [44], super-pixel CL [45], and multiview CL [46], which efficiently capture the latent pixel relationships through differentiated learning of positive and negative sample pairs and obtain competitive performance. Moreover, to overcome the locality of convolutional neural network (CNN)-based methods, some transformer-based DSC models were put forward to learn favorable global expressions and obtained impressive results [47], [48], [49]. Besides, by introducing an unfolding alternative direction of multipliers (UNADMM) to subspace clustering for deep affinity optimization, a new DSC model was proposed [50].

Nevertheless, despite these accomplishments, current DSC methods still face several common shortcomings. First, most existing methods overemphasize the importance of spatial contents and primarily design networks around spatial structure mining, while neglecting the distinctive advantage of HSIs in their continuous spectra, which leads to nonnegligible information loss and discriminability reduction. Second, although complementary local and global expressions have been incorporated into some methods by combining CNNs with transformer models, the resulting complexity diminishes their practicability in real applications. Third, current methods insufficiently explore the underlying mechanism of self-expression learning and the essential role of coefficients, which can cause affinity loss during the fusion of coefficient matrices. Collectively, these limitations constrain their clustering performance and adaptability to complex scenes.

In view of these, this article proposed a novel local–global spectral–spatial dual DSC (LGDSC) network for HSIs. On the one hand, a dual CAE architecture was established for comprehensive discriminability mining in an unsupervised way, with a shared self-expression module embedded in the immediate part between the encoders and decoders, which makes full use of spectral and spatial features, local and global expressions to promote more precise pixel affinity learning. On the other hand, a novel selective affinity aggregation (SAA) module was designed to selectively fuse favorable and indicative coefficients induced by different features to generate a more accurate adjacency matrix. Overall, the main contributions are threefold.

1) A new DSC framework named LGDSC was proposed. It fully exploits the hidden characteristics of HSIs from multiple perspectives, including spectral and spatial information as well as local and global contents. Additionally, it comprehensively explores the latent pixel relationships through a shared self-expression module, with the learned complementary affinities adaptively fused to generate precise and robust clustering results across diverse and complex HSI scenarios.
2) To compensate for discriminative information loss, a dual CAE architecture with two parallel branches was constructed, which consists of paired encoders and decoders with a symmetrical structure for differentiated feature mining. To fully leverage the flexibility and efficiency of convolutional learning, a 1-D CAE and a 2-D CAE were developed to focus on local-spectral and global-spatial feature mining, respectively, with a small-large image pair centered on the same pixel used as the input. Moreover, a valid joint loss function, integrating data reconstruction and self-expression terms with bilateral constraints on both branches, was designed to precisely guide the network training. As a result, it produces complementary feature expressions that enable precise and comprehensive pixel affinity learning.
3) To make the most of complementary self-representation coefficients derived by differentiated features, a novel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

SAA module is put forward to precisely fuse favorable pixel affinities to generate a more indicative and clustering-friendly adjacency matrix for spectral clustering. It starts from the essential meaning of coefficients to emphasize significant relationships while weakening nonsignificant ones, thereby more accurately revealing the latent partitional pattern between pixels.

The structure of this article is arranged as follows. Section II briefly reviews subspace clustering and DSC models. Section III describes the proposed LGDSC model in detail. Section IV presents the results and associated analysis. Section V concludes the article and outlines future work.

II. Related Work

A. Subspace Clustering

Subspace clustering methods model HSIs with a union of subspaces and explore the underlying pixel affinities by linearly reconstructing each pixel with others under defined structural priors. Specifically, with HSI data itself as the overcomplete dictionary, i.e., $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{D\times n}$, where D and n denote feature dimension and pixel number, respectively, subspace clustering explores the self-representation property as in the following equation:

$$\mathbf{X} = \mathbf{X}\mathbf{C} + \mathbf{E} \tag{1}$$

where $\mathbf{C} \in \mathbb{R}^{n\times n}$ and $\mathbf{E} \in \mathbb{R}^{D\times n}$ denote the coefficient matrix and error matrix, respectively. Unfortunately, the above equation has infinite solutions. To effectively reduce the solution space to acquire a stable result, various prior constraints are applied to $\mathbf{C}$ to enable it to capture the desired structures, as in the following equation:

$$\arg\min_{\mathbf{C}} \|\mathbf{C}\|_p + \lambda\|\mathbf{X} - \mathbf{X}\mathbf{C}\|_F^2 + \alpha S(\mathbf{C}) \tag{2}$$

where $\|\cdot\|_p$ denotes the ℓ_p-norm, which is the spectral term and generally comprises $\|\cdot\|_1, \|\cdot\|_*$ and $\|\cdot\|_F^2$, referring to the SSC [22], LRSC [25] and collaborative subspace clustering (CSC) models [30], respectively, $S(\cdot)$ means the spatial term, with λ and α being regularization parameters to adjust their balance. We can obtain an indicative $\mathbf{C}$ by solving (2) with alternative direction of multipliers (ADMM) and construct an adjacency matrix $\mathbf{W} \in \mathbb{R}^{n\times n}$ as in the following equation:

$$\mathbf{W} = |\mathbf{C}| + |\mathbf{C}^T|. \tag{3}$$

Then, by applying spectral clustering to $\mathbf{W}$, a meaningful clustering result can be obtained [22], [23].

To more effectively mine HSI properties, a variety of enhanced subspace clustering methods have been developed. These include five main types: 1) spectral–spatial-based; 2) multiview feature-based; 3) kernel-based; 4) dictionary learning-based; and 5) GCR-based models [9].

The spectral–spatial-based methods generally explore the spectral–spatial duality of HSIs in the self-representation process, as in (2) [22], [23], or inducing spatial information through filtering [51] or affinity propagation [52] to promote piecewise smoothness of $\mathbf{C}$, thus inducing a more accurate result. In addition, by exploring the higher order relationships and structural priors, the strength of spatial correlations is further enhanced [26]. Moreover, various super-pixel-based models were developed to overcome the shortcomings of pixel-based models in spatial neighborhood modeling and unbearable computational complexity [29]. The multiview feature-based methods make full use of complementary information found in multiview features and transform the single self-expression problem in (2) to a multiple joint self-expression problem, thereby improving the clustering performance, such as multiview sparse LRSC (SLRSC) [53] and multiview kernel SLRSC [31]. The kernel-based methods fully consider the nonlinearity of HSIs and convert self-expression from the original feature space to kernel Hilbert space through customized kernels to reduce the systematic error of linear models, such as kernel SSC [28] and kernel SLRSC [54]. The dictionary-based methods learn more indicative and expressive dictionaries for self-representation, thereby inducing more clustering-friendly coefficient matrices, such as local adaptive subdictionary collaborative representation clustering (CRC) [30] and adaptive dictionary learning based SSC [55]. The GCR-based methods extend affinity mining from Euclidean space to nonEuclidean space with GCR to better learn geometric information and showed a large potential [32], [56], such as efficient graph convolutional subspace clustering (EGCSC) [46] and graph convolutional optimal transport (GCOT) [57]. Nevertheless, despite these advances across all categories, subspace clustering methods remain constrained by their shallowness and linearity, which limit their overall clustering performance.

B. Deep Subspace Clustering

DSC is a natural extension of subspace clustering from the shadow models to deep models, which perfectly integrates the advantages of subspace clustering and deep learning in structure modeling and feature engineering. The hierarchical learning scheme of deep neural networks (DNNs) enables DSC to capture more discriminative features and explore nonlinear pixel affinities to yield more accurate clustering results. As a recently emerging field in HSI clustering, DSC has attracted increasing attention and demonstrated clear advantages over traditional methods [35], [36], [37].

DSC follows the basic framework of subspace clustering while introducing unsupervised DNNs to extract high-level semantic expressions to substitute hand-crafted features for self-expression learning by minimizing data reconstruction error, and integrates feature extraction and self-expression into a joint optimization problem. More specifically, a DSC model typically integrates an autoencoder for feature engineering, composed of an encoder and a decoder, with a fully connected self-expression module for affinity mining [35], [36], [37]. Then, a joint loss function enables unified optimization, as in the following equation:

$$\ell(\Theta, \mathbf{C}) = \beta\left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_F^2 + \gamma\|\mathbf{C}\|_p + \lambda\|\mathbf{E}_{\Theta_e} - \mathbf{E}_{\Theta_e}\mathbf{C}\|_F^2 \tag{4}$$

where $\hat{\mathbf{X}}$ denotes the reconstructed data, $\mathbf{E}_{\Theta_e}$ is the encoding feature, and Θ refers to the hyperparameters for the autoencoder composed of encoder parameter Θ_e and decoder

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

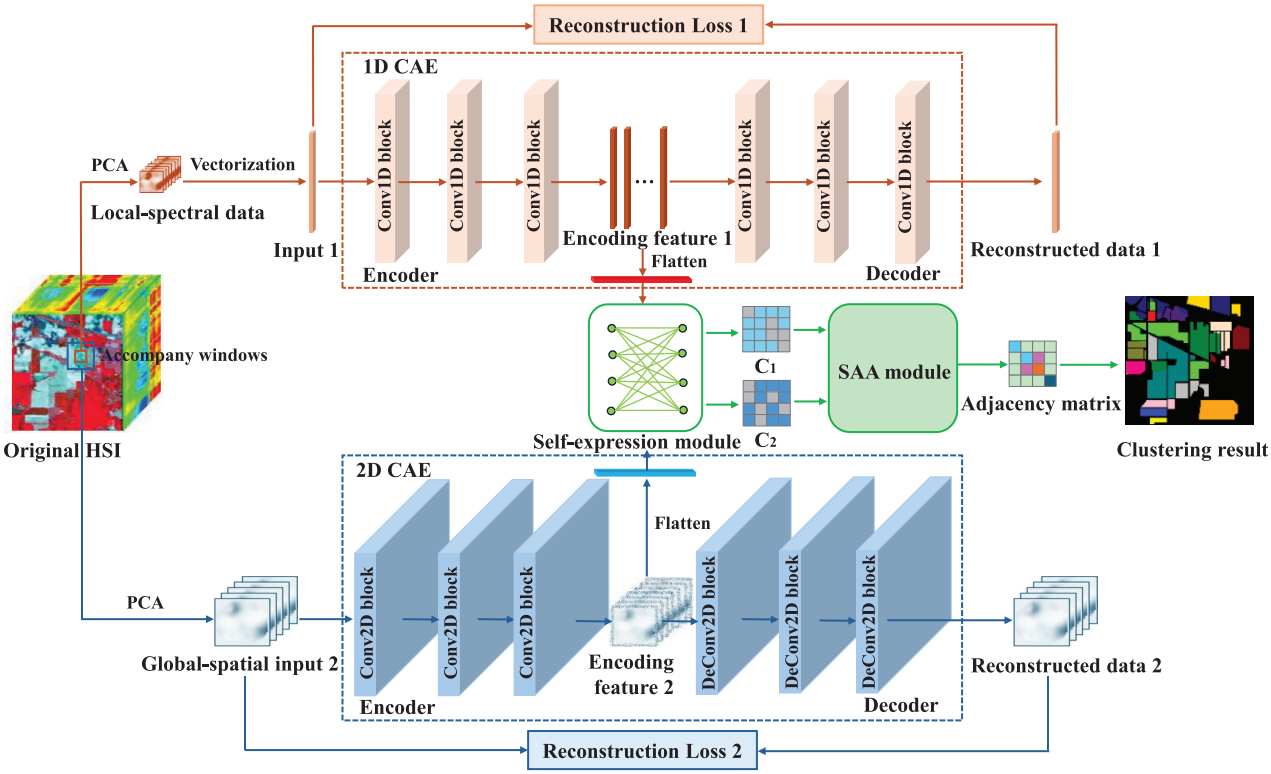


Fig. 1. Overall architecture of the LGDSC model. It adopts a typical dual CAE architecture, with 1-D CAE and 2-D CAE focusing on local-spectral and global-spatial feature mining, respectively. The learned features are fed into a shared self-expression module for unified self-representation, with a joint loss function used. Then, the learned coefficient matrices are precisely fused through the SAA module to induce a more complete and indicative adjacency matrix. By performing spectral clustering, a meaningful clustering result can be generated.

parameter Θ_d, with β and γ being tradeoff parameters. Subsequently, the learned coefficient matrix $\mathbf{C}$ is used to construct the adjacency matrix as in (3). By applying spectral clustering to it, the final clustering result can be produced [35], [36], [37].

Based on this, extensive advanced DSC methods have been developed. These methods can be divided into five main kinds based on the differences in their underlying principles: 1) convolution-based; 2) GCR-based; 3) transformer-based; 4) multiview-feature-based; and 5) CL-based models.

The convolution-based methods employ convolutions to hierarchically extract expressive spectral and spatial features in defined receptive fields for self-expression learning, such as L-CAE [38] and 3D-ACAE [39]. These methods offer strong description ability for local context, which leads to improved accuracy. The GCR-based methods embed GCR into CAEs to learn geometric information in nonEuclidean space to improve feature robustness, thus delivering competitive performance, such as robust low-rank DSC with GCR [58] and multiview GCR-DSC [59]. The transformer-based methods introduce various transformers to overcome the locality of convolutions to learn favorable global expressions for self-expression, such as transformer-based dynamic fusion clustering (TDFC) [60] and self-supervised double-structure transformer (SDST) [47]. The multiview-feature-based methods enrich feature extraction and knowledge mining by fusing complementary information from different views to extend single-view DSC to multiview joint DSC, such as contrastive multiview DSC based on GCR (CMSCGC) [59] and enhanced multiview CL-based clustering (EMCCV) [46], which further improves clustering accuracy. Finally, the CL-based methods fully exploit the potential of differentiated learning of positive and negative sample pairs to efficiently capture the underlying pixel affinities, such as two-branch CL [44] and super-pixel CL [45]. Although DSC methods have achieved notable accomplishments, they are still troubled by some common bottlenecks, such as underutilization of spectral information and insufficient exploration of the underlying mechanism of self-expression, which degrades their clustering performance.

III. LGDSC Network

This section presents a new LGDSC model that achieves high-performance clustering by fully exploiting the hidden characteristics of HSIs. Within a dual CAE framework, LGDSC extracts discriminative high-level local-spectral and global-spatial semantic features separately, with their complementary information adaptively fused through joint self-representation and SAA. This process produces an indicative and complete adjacent matrix, leading to more accurate clustering results.

A. Overall Architecture

The overall architecture of LGDSC is depicted in Fig. 1. As illustrated, LGDSC has a neat and symmetrical structure, with the dual CAE as its skeleton. It consists of two parallel branches and includes three main components: dual encoders, dual decoders, and a shared immediate part. A 1-D CAE is employed for spectral feature extraction, with a 2-D CAE used for spatial feature mining. To support simultaneous

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

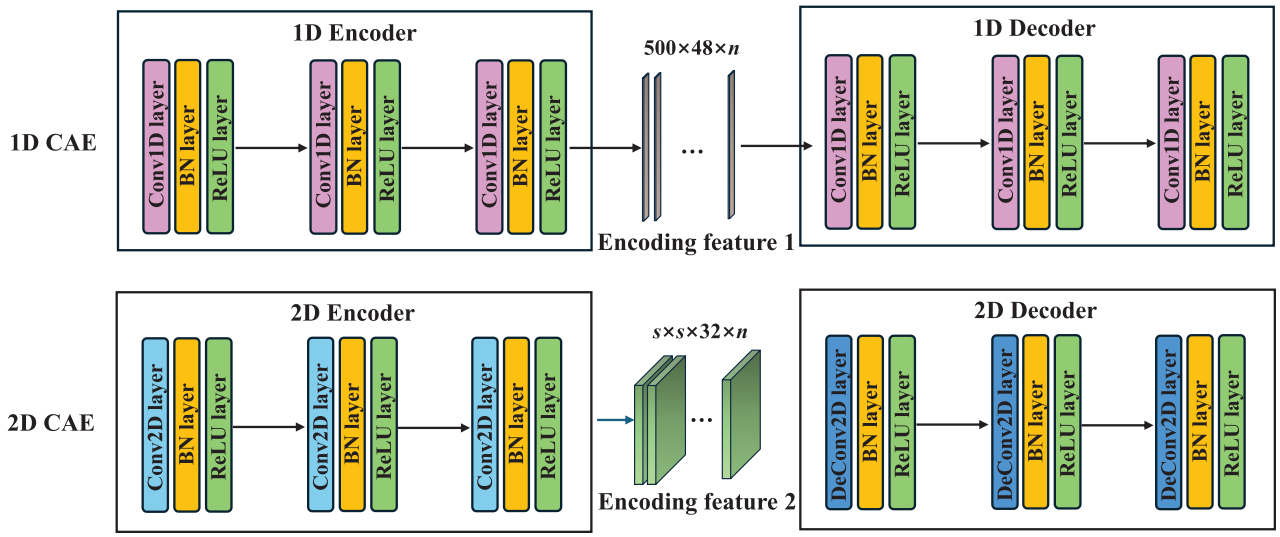


Fig. 2. Diagram of dual CAEs.

learning of local and global expressions, the framework uses an accompanying input image pair with small and large spatial ranges for each target pixel. In this way, LGDSC makes full use of the discriminative information contained in HSIs and comprehensively exploits the latent pixel relationships in multiple ways, which is conducive to improving feature expressivity and affinity completeness.

Specifically, the encoder consists of a three-layer 1-D CNN and a three-layer 2-D CNN, which employ consecutive 1-D and 2-D convolutions to deeply mine expressive and robust local-spectral and global-spatial features. The learned features are fed into a shared self-expression module to learn the latent pixel affinities. This is realized by fully connected layers that linearly reconstruct each pixel with all pixels under a defined constraint. As a result, two differentiated coefficient matrices can be produced. Subsequently, they are selectively fused by the SAA module to generate a more complete and indicative coefficient matrix. Ultimately, a fused coefficient matrix is used to construct the adjacency matrix as in (3) to perform spectral clustering, thereby yielding the final clustering result.

To guarantee the quality of encoding features, an auxiliary decoder is constructed. It has a structure symmetrical to the encoder, including a three-layer 1-D CNN to restore the local-spectral data and a three-layer 2-D CNN to restore the global-spatial data. Using the deepest encoding features, the decoder generates reconstructed data that preserves structural similarity to the inputs. By minimizing data reconstruction errors, the network can deeply mine discriminative local-spectral and global-spatial feature expressions in an unsupervised way.

B. Dual CAEs

As is well known, most of the current DSC methods overemphasize the importance of spatial pattern mining, with the network designed around a single preference, but ignore the hidden value contained in the continuous spectra. As a result, certain discriminability loss in the feature engineering process is inevitable. Hence, a more reasonable feature mining framework should be explored. For this purpose, an efficacy dual CAE architecture is constructed to make the most of the spectral–spatial and local–global contents contained in HSIs to generate more discriminative features that directly contribute to the improvement of clustering performance.

The dual CAEs consist of two parallel branches, with each branch dedicated to a specific task, as shown in Fig. 2. Since spatially adjacent pixels often share the same material composition and spectral signatures, while distant pixels provide complementary contextual information, a natural strategy is to link local features with spectral mining and global features with spatial mining. This collaborative learning approach enhances the discriminability and robustness of spectral features while improving the expressivity and richness of spatial features. Accordingly, a 1-D CAE and a 2-D CAE are constructed to focus on local-spectral and global-spatial feature extraction, respectively. Notably, the two autoencoders share a similar structure, with each basic block composed of a convolutional layer, a batch normalization (BN) layer, and a rectified linear unit (ReLU) activation layer, while differing slightly in their components. The 1-D CAE uses 1-D convolutions to build its encoder and decoder (*En*1D and *De*1D), whereas the 2-D CAE employs 2-D convolutions and deconvolutions to form its encoder and decoder (*En*2D and *Dn*2D). Additionally, an accompanying image pair, consisting of small and large neighborhoods centered on each target pixel, is used as the input for dual CAEs to provide the primary data for collaborative feature mining.

Specifically, for the 1-D CAE, a small window at a size of 5×5 is opened for each target pixel, with principal component analysis (PCA) employed to acquire the most important 20 spectral components, thereby generating the local-spectral data $\mathbf{X}_{1D} \in \mathbb{R}^{5\times5\times20\times n}$, where n denotes the number of pixels. Then, the image patch is vectorized to align with the working mechanism of a 1-D CNN, yielding an input $\mathbf{I}_{1D} \in \mathbb{R}^{500\times n}$. For the 2-D CAE, a large window at a size of $s \times s$ is opened for each target pixel, using PCA to reduce the feature dimension to 5 to generate the global-spatial input $\mathbf{I}_{2D} \in \mathbb{R}^{s\times s\times5\times n}$, where s depends on the image

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

characteristics. Subsequently, $\mathbf{I}_{1D}$ and $\mathbf{I}_{2D}$ are fed into *En*1D and *En*2D to deeply mine robust and discriminative local-spectral and global-spatial expressions through hierarchical feature processing. Herein, the kernel size and stride size are set to 3 and 1 for both 1-D and 2-D convolutions. To enrich feature expressions, 24, 32, 48 and 24, 24, 32 kernels are deployed for the first, second, and third convolutional layers of *En*1D and *En*2D, with their encoding features of ith layer denoted as $\mathbf{Z}_{1D}^{i} \in \mathbb{R}^{500\times k\times n}$ and $\mathbf{Z}_{2D}^{i} \in \mathbb{R}^{s\times s\times g\times n}$, where k and g stand for the number of kernels. In addition, to ensure sufficient network depth, the same padding operation is applied to preserve the feature size. As a result, two differentiated encoding features are produced, i.e., $\mathbf{Z}_{1D} \in \mathbb{R}^{500\times 48\times n}$ and $\mathbf{Z}_{2D} \in \mathbb{R}^{s\times s\times 32\times n}$, which are the basis for decoding and self-expression learning.

To ensure that the encoding features successfully capture the intrinsic structural properties of the data, symmetrical decoders are constructed, i.e., *Dn*1D and *Dn*2D, to recover the input data as much as possible based on the deepest encoding features $\mathbf{Z}_{1D}^{3}$ and $\mathbf{Z}_{2D}^{3}$. To reconstruct the local-spectral and global-spatial data, 1-D convolutions and 2-D deconvolutions with a kernel size of 3 and a stride of 1 are used for *Dn*1D and *Dn*2D, with their hierarchical kernel number assigned as 48, 32, 24 and 32, 24, 24, respectively. Similarly, the same padding operation is used to maintain the feature size. Accordingly, accompanying outputs are generated through progressive reconstruction, i.e., $\hat{\mathbf{I}}_{1D} \in \mathbb{R}^{500\times n}$ and $\hat{\mathbf{I}}_{2D} \in \mathbb{R}^{s\times s\times 5\times n}$. By minimizing data reconstruction error, dual CAEs effectively learn expressive and reversible features for each target pixel in an unsupervised way.

Based on the streamlined architecture, dual CAEs are capable of fully exploiting the spectral–spatial and local–global information of HSIs, thus producing complementary and differentiated features for self-expression learning. This is conducive to improving the discriminability of features and supporting thorough exploration of pixel relationships to enhance the completeness of affinities. As a result, the model delivers more accurate clustering results.

C. SAA Module

With the help of the differentiated local-spectral and global-spatial features learned by dual CAEs, i.e., $\mathbf{Z}_{1D}$ and $\mathbf{Z}_{2D}$, two complementary coefficient matrices $\mathbf{C}_1 \in \mathbb{R}^{n\times n}$ and $\mathbf{C}_2 \in \mathbb{R}^{n\times n}$ can be obtained through the shared self-expression module, which captures the underlying structural information of HSIs from different perspectives, with direct impact on subsequent spectral clustering. Therefore, how to effectively fuse the favorable affinities derived from $\mathbf{C}_1$ and $\mathbf{C}_2$ while screening out the interference to generate a more complete and clustering-friendly coefficient matrix is the key issue. At this point, a novel SAA module is put forward, starting from the underlying mechanism of self-representation and the essential meaning of coefficients, to precisely fuse the complementary information of $\mathbf{C}_1$ and $\mathbf{C}_2$, thus promoting more accurate pixel partitions.

The SAA module has a typical three-branch structure, as depicted in Fig. 3. Specifically, an elementary branch is built to violently aggregate pixel affinities provided by $\mathbf{C}_1$ and $\mathbf{C}_2$ through an element-wise addition, which adopts a nonselective

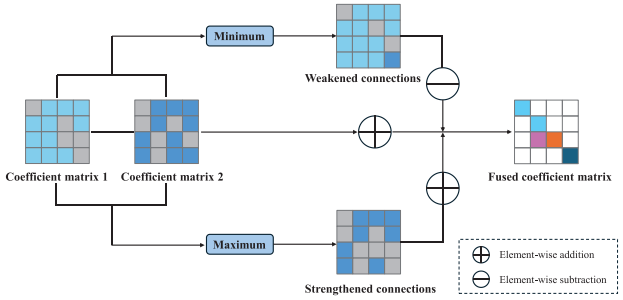


Fig. 3. Structure of the SAA module.

fusion strategy to guarantee the completeness of pixel affinities. However, a large amount of redundancy and interference are inevitably included in this violent fusion process, resulting in nonnegligible degradation of subsequent spectral clustering. To compensate for this shortcoming, two selective fusion branches are well designed, with one branch highlighting the salient pixel affinities and the other branch reducing negative effects caused by nonsalient relationships. By this means, reasonable adjacencies are effectively strengthened, while incorrect adjacencies are rectified, thereby ensuring that each pixel can be separated into the correct cluster.

According to the principle of self-representation, larger coefficients indicate stronger affinities between associated pixels. Since the coefficient matrices are derived from different features, some differentiated connections are expected in the corresponding similarity graph. To integrate their advantages in pixel adjacency description, the connections between pixels from the same class are reinforced to ensure that they can be segmented into the same group. To achieve this reinforcement, an element-wise comparison strategy is designed to select the larger coefficient at each position, thereby constructing a salient coefficient matrix, i.e., $\mathbf{C}_{\max}$, as formulated in the following equation:

$$\mathbf{C}_{\max} = \max\left(\mathbf{C}_1, \mathbf{C}_2\right). \tag{5}$$

By adding $\mathbf{C}_{\max}$ to the sum of $\mathbf{C}_1$ and $\mathbf{C}_2$, their positive contributions can be effectively amplified.

Similarly, smaller coefficients indicate weak correlations between associated pixels, suggesting that they belong to different classes and are likely to represent redundant or interfering information. Generally, these weak connections offer minimal insight into latent pixel relationships, instead exerting negative effects on graph cutting. Hence, the smaller coefficient at each position is extracted to form an interference coefficient matrix, i.e., $\mathbf{C}_{\min}$, as in the following equation:

$$\mathbf{C}_{\min} = \min\left(\mathbf{C}_1, \mathbf{C}_2\right). \tag{6}$$

By subtracting $\mathbf{C}_{\min}$ from the sum of $\mathbf{C}_1$ and $\mathbf{C}_2$, the invalid connections are weakened, and their negative effects are eliminated, thereby reducing mis-segmentations.

By integrating the results derived from all three branches, a more informative coefficient matrix $\mathbf{C}$ can be produced

$$\mathbf{C} = \mathbf{C}_1 + \mathbf{C}_2 + \mathbf{C}_{\max} - \mathbf{C}_{\min}. \tag{7}$$

Based on $\mathbf{C}$, a more precise adjacency matrix can be derived as in (3). Through spectral clustering, a meaningful clustering result is yielded.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

In this way, the SAA module fully leverages the underlying mechanism of self-expression to selectively aggregate the learned affinities, using larger coefficients to strengthen salient relationships and smaller ones to weaken nonsalient relationships. As a result, favorable pixel adjacencies are enhanced while interference is eliminated, which facilitates the revelation of the intrinsic structure of HSIs and brings about a notable improvement in clustering accuracy.

D. Joint Loss Function With Bilateral Constraints

Considering that the LGDSC model encompasses two parallel branches for different tasks, it is necessary to constrain them simultaneously to guide feature learning in the training process. To this end, by fully considering their common characteristics, the loss function of traditional DSC models is extended to an effective joint loss function with bilateral constraints, which reuses the definition of each constraint without introducing additional regularization forms. In this manner, complementary tasks are effectively balanced, while maintaining the simplicity of the loss function, making it easy to solve using the same strategy as traditional DSC methods.

For the local-spectral branch, a common loss function is defined, which consists of two parts: reconstruction loss and self-expression loss. The reconstruction loss is defined as the total difference between the input data $\mathbf{I}_{1D}$ and reconstructed data $\hat{\mathbf{I}}_{1D}$ induced by the decoder $Dn1D$, as in the following equation:

$$\text{loss}_{\text{DR}_{1D}} = \alpha \|\mathbf{I}_{1D} - \hat{\mathbf{I}}_{1D}\|_F^2 \tag{8}$$

where α is the regularization parameter. Based on the deep-seated encoding feature $\mathbf{Z}_{1D}$, the self-expression loss can be modeled as the integration of a structural prior term and data fidelity term, as in the following equation:

$$\text{loss}_{\text{SE}_{1D}} = \beta \|\mathbf{Z}_{1D} - \mathbf{Z}_{1D}\mathbf{C}_1\|_F^2 + \gamma \|\mathbf{C}_1\|_F^2 \tag{9}$$

where β and γ are trade-off parameters. Here, the Frobenius norm is used to promote an efficient solution of $\mathbf{C}_1$. By integrating these two parts, the loss function for the local-spectral branch can be obtained, as in the following equation:

$$\text{loss}_{LS} = \alpha \left\|\mathbf{I}_{1D} - \hat{\mathbf{I}}_{1D}\right\|_F^2 + \beta \|\mathbf{Z}_{1D} - \mathbf{Z}_{1D}\mathbf{C}_1\|_F^2 + \gamma \|\mathbf{C}_1\|_F^2. \tag{10}$$

A similar loss function can be defined for the global-spatial branch, which also consists of reconstruction loss and self-expression loss, as formulated in the following equation:

$$\text{loss}_{\text{GS}} = \alpha \left\|\mathbf{I}_{2D} - \hat{\mathbf{I}}_{2D}\right\|_F^2 + \beta \|\mathbf{Z}_{2D} - \mathbf{Z}_{2D}\mathbf{C}_2\|_F^2 + \gamma \|\mathbf{C}_2\|_F^2. \tag{11}$$

By integrating the loss functions of these two branches, a final joint loss function can be obtained, as in the following equation:

$$\text{loss}_J = \text{loss}_{LS} + \text{loss}_{\text{GS}}. \tag{12}$$

By defining these bilateral constraints, the joint loss function effectively guides the learning process of both local-spectral and global-spatial branches, while maintaining unified form and simple composition. This leads to more discriminative features and more indicative coefficient matrices, thereby facilitating higher clustering performance.

IV. Experiment and Analysis

This section presents extensive experimental results conducted on four public HSI datasets to validate the efficacy of LGDSC, with the clustering performance of different methods analyzed and comprehensively compared in detail.

A. Experimental Setting

1) Experimental Datasets: To verify the effectiveness of LGDSC, four widely used HSI datasets were selected: Indian Pines (IP) image, Pavia University (PU) image, Houston University (HU) image, and Xuzhou (XZ) image. These scenes exhibit varied and distinct land cover characteristics and provide a solid basis for comprehensive performance evaluation.

The IP image was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over an agricultural area in Indiana, USA. This scene has a size of 145 × 145 × 224, with a spatial resolution of 20 m. After removing 24 noisy bands, 200 bands were retained for clustering analysis. This scene includes 16 classes, many of which are similar subclasses, making clustering particularly challenging. For this moderate-sized image, the complete scene was used in the experiments.

The PU image was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) over PU, Italy. After discarding noisy bands, it has a size of 610 × 340 × 103, with a high spatial resolution of 1.3 m, covering nine land cover classes. To ensure computational efficiency, following most prior works [41], [42], [43], a representative subscene located at [151–350, 101–200] was selected for clustering evaluation, containing eight classes.

The HU image was captured by the National Center for Airborne Laser Mapping (NCALM) sensor over HU and its surrounding areas, USA. This scene is very large, with a size of 349 × 1905 × 144. It has a spatial resolution of 2.5 m and includes 15 classes. Following [41], [42], and [43], a representative subscene located at [1–349, 1–680] was chosen, containing 12 classes. Unlike the other datasets, this scene features numerous small objects with fragmented and complex distributions, which makes clustering very difficult.

The XZ image was collected by the HySpex Imaging Spectrometer (HySpex) over a suburban mining area in Xuzhou city, Jiangsu province, China. It spans a spectral range of 415–2508 nm with a high spatial resolution of 0.73 m, including 436 spectral bands. For clustering analysis, a subscene located at [1–100, 1–260] was selected, which contains five representative classes: crops, bare land, red tiles, trees, and cement [61]. The complex spatial distribution of these ground objects leads to a large difficulty in clustering.

2) Evaluation Metrics: To comprehensively assess the clustering performance of different methods, both qualitative and quantitative results were given. Four common metrics were employed for accuracy assessment, i.e., producer's accuracy (PA), user's accuracy (UA), overall accuracy (OA), normalized

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NETWORK CONFIGURATIONS FOR DUAL CAE

Branch	Layer No.	Operation	Kernel size	Kernel number	Stride size	Padding	Activation
1D CAE	Encoding layer 1	1D convolution	3	24	1	Same	BN + ReLU
	Encoding layer 2	1D convolution	3	32	1	Same	BN + ReLU
	Encoding layer 3	1D convolution	3	48	1	Same	BN + ReLU
	Decoding layer 1	1D convolution	3	48	1	Same	BN + ReLU
	Decoding layer 2	1D convolution	3	32	1	Same	BN + ReLU
	Decoding layer 3	1D convolution	3	24	1	Same	BN + ReLU
2D CAE	Encoding layer 1	2D convolution	3×3	24	1×1	Same	BN + ReLU
	Encoding layer 2	2D convolution	3×3	24	1×1	Same	BN + ReLU
	Encoding layer 3	2D convolution	3×3	32	1×1	Same	BN + ReLU
	Decoding layer 1	2D deconvolution	3×3	32	1×1	Same	BN + ReLU
	Decoding layer 2	2D deconvolution	3×3	24	1×1	Same	BN + ReLU
	Decoding layer 3	2D deconvolution	3×3	24	1×1	Same	BN + ReLU

mutual information (NMI), and Kappa statistic [41], [42], [43], with PA and UA used to indicate the single-class precision, and OA, NMI, Kappa used to indicate the total precision. For all metrics, larger values within the range of 0–1 correspond to higher clustering accuracy and performance quality. The Hungarian algorithm was used to automatically align clustering labels with real labels in the ground truth [41].

3) Comparative Methods: To verify the efficacy of the proposed LGDSC model, seven baselines were used for comparison. These include both popular subspace clustering methods and SOTA DSC methods. Among them, low-rank representation (LRR) [62] and efficient dense subspace clustering (EDSC) [63] are two classical subspace clustering methods, EGCSC [45] and GCOT [57] are two advanced subspace clustering methods, and GR_RSCNET [41], HyAE [42], and UNADMM [50] are three SOTA DSC methods. The source codes were used for all benchmarks, and the parameters were set as recommended.

4) Implementation Details: The LGDSC model takes dual CAE as its skeleton and follows an unsupervised learning framework. The detailed network configurations are listed in Table I. This model was implemented in the TensorFlow framework (version 2.0.0) using Python and trained from scratch by the Adam optimizer. The learning rate and maximum epoch were unified as 0.0002 and 100 for all four datasets, with hyperparameters of the joint loss set as $\alpha = 1, \beta = 100$ and $\gamma = 0.001$, based on the common grid searching strategy. The details are given in Section IV-F. In addition, all experiments were performed on a server with an Intel Xeon W-2245 CPU and 64 GB of RAM. Since some of the baselines are classical subspace clustering methods, for fair comparison and comprehensive testing, both the CPU and GPU were used, with the GPU equipped with a NVIDIA Quadro RTX 5000 with a memory of 16 GB. All methods were executed on the CPU across the four datasets for efficiency comparison, where four DSC methods were also executed on the GPU for PU and HU images to enable further analysis where sufficient computing power was available.

B. Experimental Results

The clustering results for each dataset derived by different methods were comprehensively compared. Both visual and quantitative results are presented to evaluate performance.

1) Experiment on the IP Dataset: Fig. 4 and Table II provide the experimental results of different methods on the IP dataset. The results clearly indicate the advantages of DSC methods over traditional subspace clustering methods and the significant superiority of LGDSC to the other models. Notably, most methods showed reduced efficacy in this complex scenario, whereas LGDSC well distinguished the majority of classes. Accordingly, it achieved the best result for this scene, with a much better visual map and much higher precision, which suggests its strong capability in dealing with highly similar ground objects.

Specifically, LRR and EDSC performed poorly on this dataset, yielding low OAs of only about 30%. As observed in Fig. 4, there were a large number of misclassifications in the visual maps, with similar classes seriously confused, especially for corn, corn-notill, and corn-mintill. As a result, the PAs and UAs for these classes were only around 20%. The reason for this cloud be that these methods focus on spectral exploitation in self-expression learning without spatial assistance, resulting in limited discriminability. EGCSC and GCOT significantly improved clustering performance using GCR to promote spectral–spatial collaboration and geometrical information mining in the nonEuclidean space. The identification precision of most classes was largely improved. Especially for EGCSC, it performed competitively for many classes, such as corn-notill, corn, grass-pasture-mowed, soybean-mintill, and wheat, to yield a promising OA of 60.45%. However, it displayed obvious weakness for alfalfa, oats, and stone-steel-towers, with the PAs being zero. GCOT delivered impressive performance for stone-steel-towers using an optimal transport strategy, but lost efficacy for grass-pasture and soybean-clean. Overall, due to the inherent shortcomings of shallow models in discriminability mining and nonlinear affinity learning, these traditional methods failed to yield satisfactory results for this scene.

By comparison, the DSC methods effectively overcame these obstacles to show clear advantages over most subspace clustering methods by subtly combining deep learning with subspace clustering. By learning robust high-level semantic features through a residual CAE, GR_RSCNET significantly outperformed LRR and EDSC to yield a cleaner map with fewer misclassifications. For corn-notill and grass-trees, the PAs were improved by over 40%, with the UAs also obviously

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

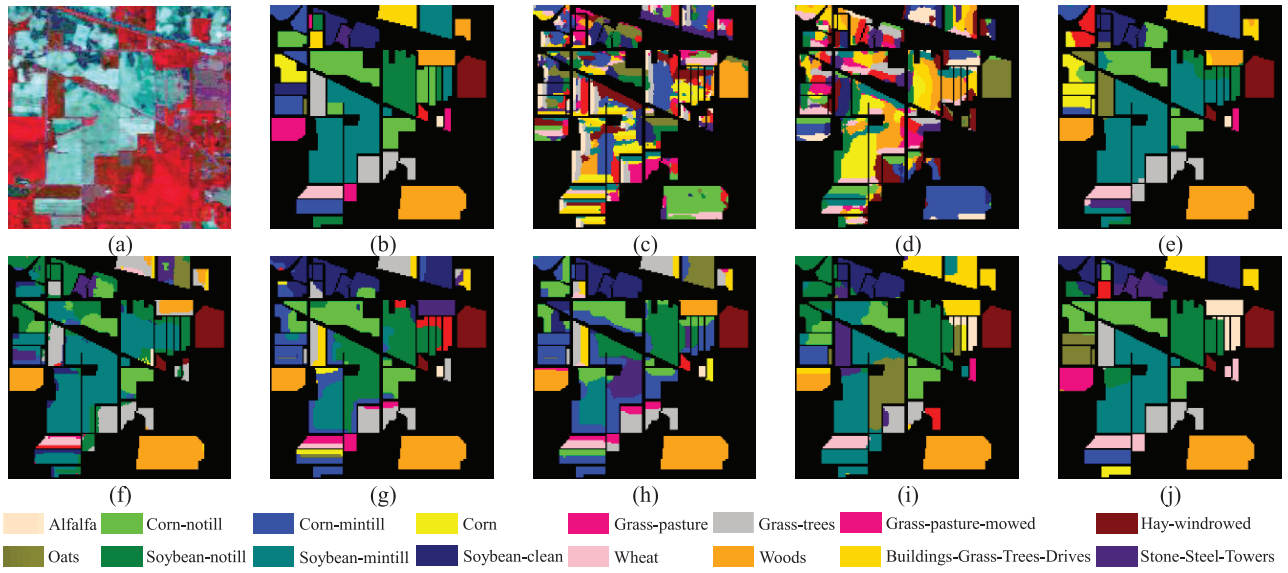


Fig. 4. Clustering results derived by different methods for the IP image. (a) False color image, (b) ground truth, (c) LRR, (d) EDSC, (e) EGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) LGDSC.

improved. However, it experienced serious confusion on highly similar classes because of information underutilization. As a result, it yielded a limited OA of 48%. HyAE performed much better by mining higher order relationships based on a hypergraph. Compared to GR_RSCNET, it obtained large gains of 19.05% and 5.89% in PA and UA, respectively, for woods. While for alfalfa, the PA was improved from 0% to 100%. Consequently, it obtained an obvious gain of 3.62% in OA. UNADMM further improved the clustering performance to yield a higher OA of 53.68% by deeply optimizing the coefficient matrix with unfolded ADMM. This model obtained over 33% and 22% gains for wheat in PA and UA, respectively. While for oats, it improved the PA from 0% to 100%. Nevertheless, due to the ignorance of spectral contributions and inadequate exploration of the self-expression mechanism, these DSC models were troubled by degraded discriminability and limited adaptability when dealing with highly similar objects.

In contrast, the proposed LGDSC model effectively tackled these problems by making full use of local-spectral and global-spatial information contained in HSIs through dual CAE while precisely fusing complementary pixel affinities through the SAA module. It achieved the best result on this dataset, with the highest OA of 67.41%, NMI of 72.90%, and Kappa of 0.6369, exhibiting noticeable improvements over the other methods. Specifically, LGDSC significantly surpassed the second-best method, with 6.96%, 7.19%, and 8.92% gains in terms of OA, NMI, and Kappa, respectively. Compared with the other three DSC methods, i.e., UNADMM, HyAE, and GR_RSCNET, LGDSC presented an overwhelming advantage, with 13.73%, 15.79%, and 19.41% increments in OA, obviously indicating its efficacy. Furthermore, it well distinguished highly similar classes to yield the optimal or suboptimal precision for most classes. For example, for grass-pasture, it obtained noteworthy improvements in both PA and UA. Especially for hay-windrowed, oats, and wheat, it performed perfectly, with the PAs being 100% and the UAs also at a high level. Moreover, LGDSC achieved a much smoother and cleaner clustering map, with greatly reduced misclassifications and within-class noise, which was closest to the ground truth. However, some limitations still remained for alfalfa, corn, and grass-pasture-mowed, which will be addressed in our next effort.

2) Experiment on the PU Dataset: Fig. 5 and Table III give the clustering results on the PU dataset. As observed, DSC methods presented more obvious advantages over traditional subspace clustering methods, and LGDSC performed best for this scene, delivering impressive visual effects and quantitative accuracy. As can be seen, ground objects are relatively concentrated and regular in this scene, making most SOTA methods obtain promising results. Nevertheless, LGDSC achieved an obvious performance boost, with all indicators being far ahead, which further demonstrates its effectiveness.

The two classical models, i.e., LRR and EDSC, yielded poor performance for this scene again, with their OAs being less than 50%, due to the absence of spatial assistance. As shown in the clustering maps, serious misclassifications existed. The vast majority of asphalt, tree, and bricks cannot be identified, with their PAs and UAs being under 20%. Comparatively, the two advanced subspace clustering methods, i.e., EGCSC and GCOT, greatly improved the clustering accuracy using GCR and optimal transport, to yield much higher OAs of 84.42% and 91.02%. They greatly improved the identification effect of meadows, metal sheet, and bare soil, obtaining large gains over 50% in PA, with the increments in UA being also obvious. However, limited by weak discriminability and robustness of shallow features, they still lost power for the above three classes.

In contrast, the three DSC models, i.e., UNADMM, HyAE, and GR_RSCNET, effectively overcame these obstacles, with the help of deep-seated features and nonlinear affinities, to successfully distinguish these classes and yielded competitive OAs of about 95% for this scene, which suggests the superiority of DSC again. However, as they ignored spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS FOR THE IP IMAGE

Metric	Class	LRR	EDSC	EGCSC	GCOT	GR_RSCNET	HyAE	UNADMM	LGDSC
PA (%)	1	41.30	2.17	0	0	0	**100**	0	0
	2	15.97	15.13	59.31	35.50	59.38	50.84	44.26	**64.78**
	3	16.63	23.98	26.63	8.55	43.49	**64.70**	26.02	32.53
	4	28.69	20.68	**63.29**	0	0	0	0	0
	5	18.01	14.70	12.22	0.41	18.22	15.53	12.22	**65.22**
	6	32.47	31.64	63.01	88.49	73.01	66.30	44.11	**100**
	7	0	10.71	**100**	0	**100**	**100**	0	0
	8	88.28	97.28	**100**	**100**	**100**	**100**	**100**	**100**
	9	65.00	0	0	0	0	0	**100**	**100**
	10	34.67	26.85	19.75	32.10	64.51	66.67	**76.23**	**76.23**
	11	12.14	27.05	**76.54**	71.24	21.67	18.33	49.12	70.75
	12	33.39	17.54	42.50	2.02	63.07	**75.21**	72.68	32.72
	13	33.66	61.46	**100**	60.00	40.98	65.37	99.02	**100**
	14	56.92	69.17	89.17	89.01	71.78	**90.83**	71.46	71.46
	15	39.12	36.01	**76.94**	8.55	15.28	23.32	75.13	**76.94**
	16	0	48.39	0	**100**	0	0	0	**100**
UA (%)	1	3.14	0.26	0	0	0	**100**	0	0
	2	51.12	19.63	73.14	63.61	**74.25**	54.95	73.57	70.56
	3	20.56	31.24	35.13	57.72	27.08	24.04	**100**	**100**
	4	10.95	12.59	**30.30**	0	0	0	0	0
	5	12.03	15.10	**100**	6.89	33.20	28.19	**100**	**100**
	6	40.10	28.73	82.73	59.04	57.55	56.14	**100**	99.73
	7	0	0.43	**100**	0	**100**	**100**	0	0
	8	63.07	92.26	91.22	88.68	91.22	**100**	**100**	91.22
	9	2.28	0	0	0	0	0	2.41	**5.33**
	10	29.82	31.79	48.60	18.96	35.28	**63.21**	45.15	58.58
	11	45.77	75.54	61.68	5.21	80.97	**98.25**	56.72	95.59
	12	33.39	26.46	52.60	4.58	29.21	57.54	**72.19**	20.14
	13	12.32	25.35	91.11	74.09	**100**	76.57	99.02	53.95
	14	81.08	75.30	78.00	76.44	74.06	79.95	79.64	**99.67**
	15	31.78	28.54	69.06	49.25	21.77	33.08	27.83	**71.22**
	16	0	10.48	0	23.25	0	0	0	**27.19**
OA (%)		29.12	33.65	60.45	50.27	48.00	51.62	53.68	**67.41**
NMI (%)		24.05	36.33	65.71	51.80	55.60	59.21	60.31	**72.90**
Kappa		0.2842	0.2797	0.5477	0.4244	0.4275	0.4691	0.4825	**0. 6369**

contributions, these models had some defects. For instance, GR_RSCNET missed certain parts of bitumen and shadows, while HyAE and UNADMM lost efficacy for bricks and shadows. Consequently, they failed to yield satisfactory results.

By comparison, LGDSC further improved the clustering performance to yield a perfect result with all indicators being 100%, which means that it completely mastered this data. Compared with the second-best method, i.e., GR_RSCNET, it achieved large gains of 4.42%, 5.41%, and 5.32% in OA, NMI, and Kappa. Compared with the other two DSC methods, the improvements were more obvious. Furthermore, it attained noticeable PA values of 100% for all classes. It perfectly distinguished metal sheet, bricks, and shadows while most of the other methods failed on these classes. Accordingly, LGDSC delivered an accurate visual map that was fully consistent with the ground truth, which further indicates its superiority.

3) Experiment on the HU Dataset: Fig. 6 and Table IV present the clustering results on the HU dataset. From which, similar conclusions can be drawn that DSC models significantly outperformed traditional models, and LGDSC displayed an incomparable advantage over the other methods. As illustrated, due to the small size and complex distribution of ground objects, most of the comparative methods were less effective, with low accuracy. LGDSC successfully distinguished those

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

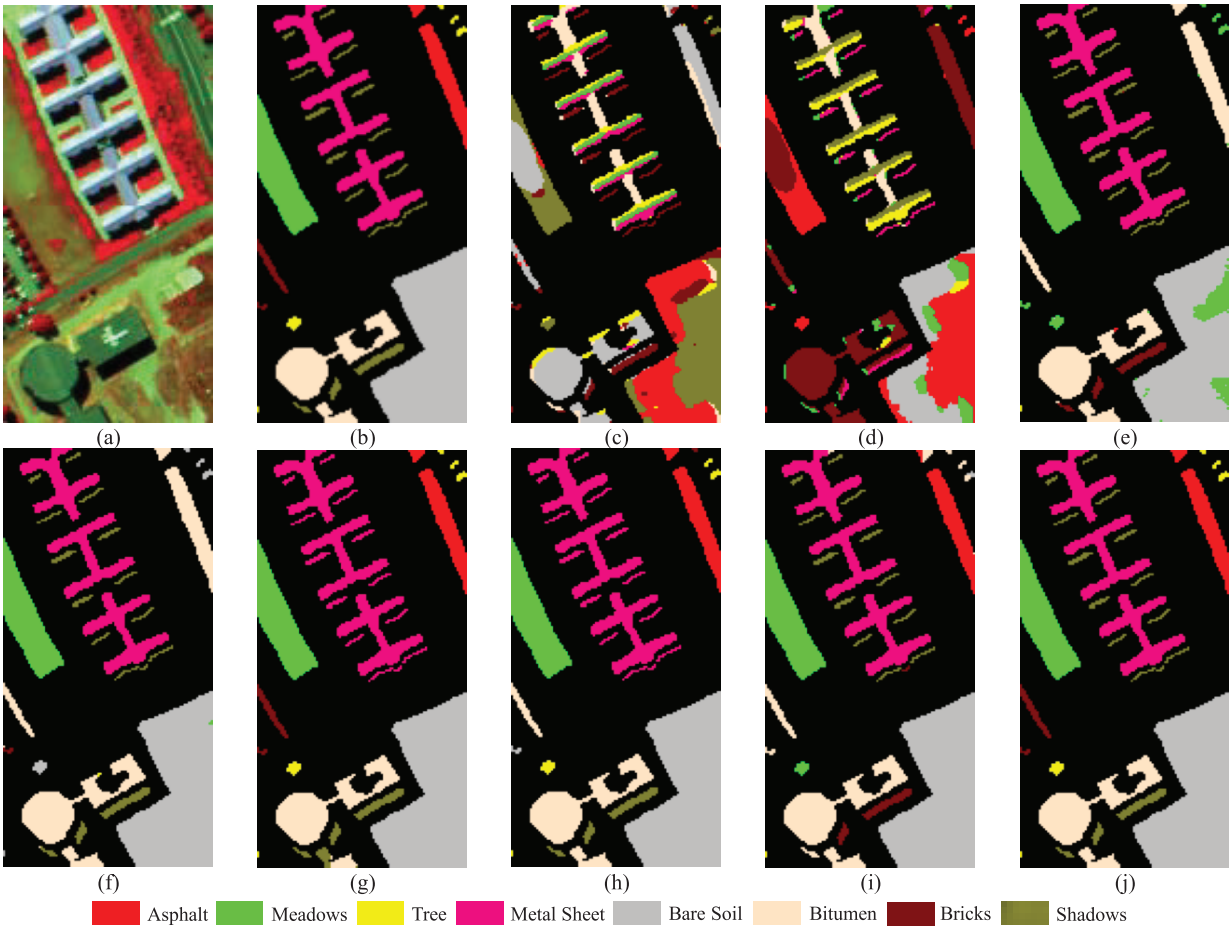


Fig. 5. Clustering results derived by different methods for the PU image. (a) False color image, (b) ground truth, (c) LRR, (d) EDSC, (e) EGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) LGDSC.

TABLE III
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS FOR THE PU IMAGE

Metric	Class	LRR	EDSC	EGCSC	GCOT	GR_RSCNET	HyAE	UNADMM	LGDSC
PA (%)	1	20.00	0.47	0	0	**100**	**100**	95.06	**100**
	2	51.04	54.17	**100**	**100**	**100**	**100**	**100**	**100**
	3	11.11	0	4.76	0	**100**	92.06	0	**100**
	4	25.32	34.75	**100**	**100**	**100**	**100**	99.85	**100**
	5	44.31	39.66	89.29	99.88	**100**	**100**	55.49	**100**
	6	81.74	93.49	99.53	99.65	94.88	**100**	**100**	**100**
	7	0.00	12.77	0	10.64	**100**	0	0	**100**
	8	86.70	66.20	59.28	99.72	40.44	40.44	57.89	**100**
UA (%)	1	14.73	0.41	0	0	**100**	**100**	**100**	**100**
	2	23.90	23.82	68.87	99.61	**100**	**100**	94.00	**100**
	3	1.63	0	**100**	0	**100**	**100**	0	**100**
	4	**100**	**100**	**100**	**100**	85.94	85.94	**100**	**100**
	5	97.75	99.90	99.95	97.59	**100**	99.57	**100**	**100**
	6	47.75	47.07	62.80	62.69	**100**	91.19	89.21	**100**
	7	0	3.09	0	**100**	**100**	0	25.92	**100**
	8	59.61	89.85	99.53	**100**	76.84	**100**	**100**	**100**
OA (%)		46.04	45.69	84.42	91.02	95.58	95.13	94.86	**100**
NMI (%)		52.47	56.91	84.01	92.33	94.59	93.88	93.90	**100**
Kappa		0.3617	0.3586	0.7968	0.8804	0.9468	0.9354	0.9223	**1.0000**

objects to yield an excellent result with few misclassifications and high-level PAs and UAs. It was the only method with an OA over 90%, which indicates its strong adaptability and promising practicability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

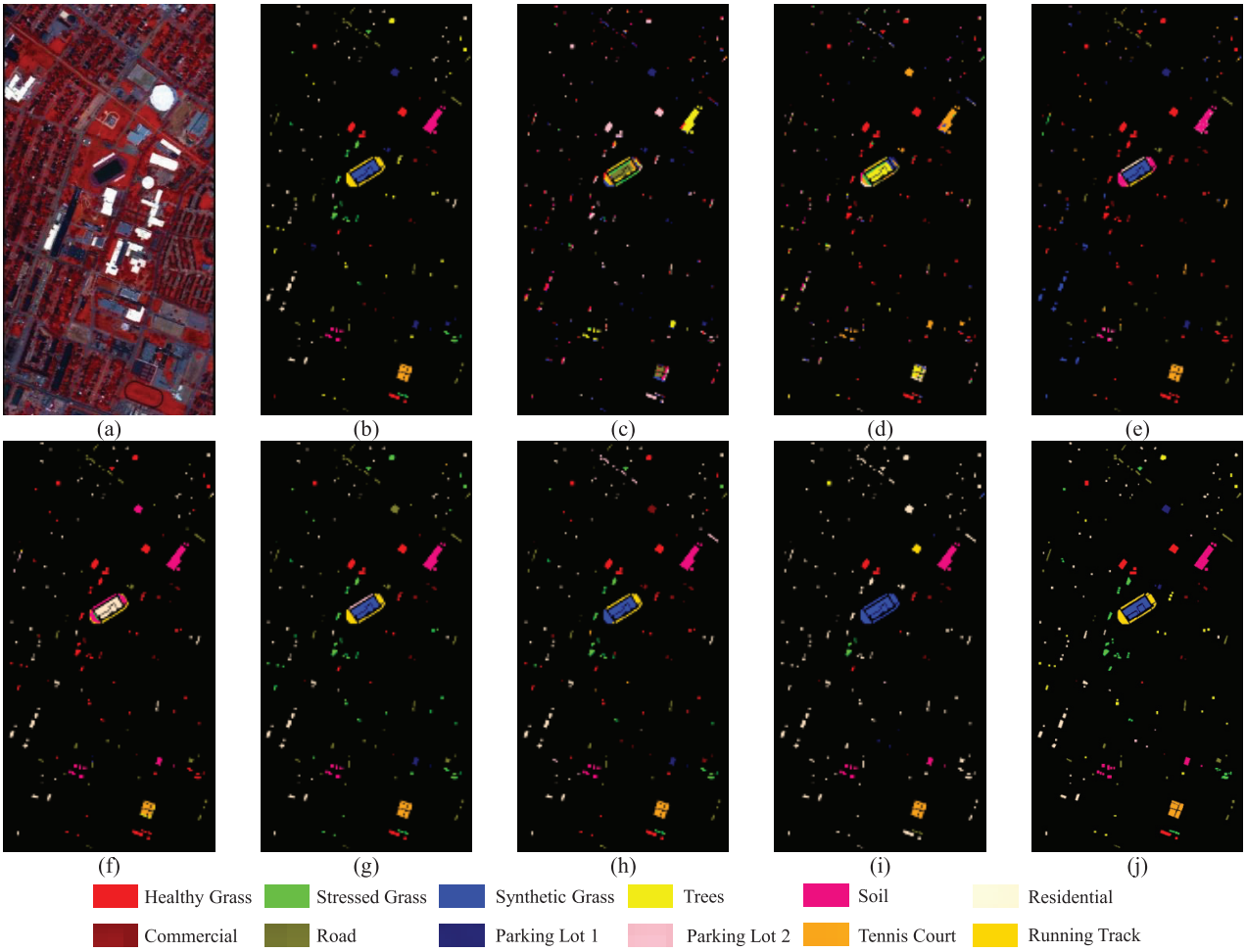


Fig. 6. Clustering results derived by different methods for the HU image. (a) False color image, (b) ground truth, (c) LRR, (d) EDSC, (e) EGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) LGDSC. (All maps are rotated 90°.)

LRR and EDSC suffered from serious misclassifications and obtained the worst performance for this scene, with their OAs being nearly 40%. EGCSC and GCOT improved the clustering performance to some degree and yielded higher OAs of about 53%. However, misclassifications in the visual maps were still serious, especially for stressed grass, trees, and parking lot 2, with their PAs and UAs being zero or near zero. GR_RSCNET and HyAE greatly improved the performance and obtained impressive results, with UNADMM being an exception. These two DSC methods effectively alleviated these problems and yielded high-level OAs of 78.53% and 83.22%. However, these methods were less effective for trees and packing lot 1, leading to unsatisfactory clustering results.

By comparison, LGDSC effectively compensated for these shortcomings and delivered the best performance for this complex scene, with much higher OA, NMI, and Kappa of 92.87%, 90.99%, and 0.9206 obtained. It achieved a remarkable improvement of 9.65% in OA over the subbest method, i.e., GR_RECNET. Compared with the other DSC methods, the advantages were more significant. It performed well for most classes to yield the optimal or suboptimal PAs and UAs. Especially for trees and running track, it attained noteworthy PAs and UAs of 98.64%, 100%, 99.10%, 100%, while the other methods lost efficacy. As a result, LGDSC achieved the best visual effect, with the clustering map being very close to the ground truth. There were clear boundaries between different classes and fewer within-class noises.

4) Experiment on the XZ Dataset: Fig. 7 and Table V give the clustering results on the XZ dataset, which are consistent with the other experimental results. As observed, DSC methods surpassed most traditional models, and LGDSC performed best to be the only method with an OA over 95%, exhibiting significant advantages over the other arts.

Specifically, LRR suffered from serious misclassifications, with a much lower OA of only 64.73%. Unexpectedly, EDSC obtained very competitive performance, yielding the second-best result. However, it failed to identify red tiles, with the PA and UA values being under 13%. EGCSC and GCOT outperformed LRR, but not EDSC, with less efficacy for red tiles and cement. In contrast, GR_RSCNET, HyAE, and UNADMM improved the clustering performance to obtain higher accuracy, which performed much better for these classes, but were troubled by a serious omission of bare land. By comparison, LGDSC effectively overcame these obstacles to deliver an accurate result for each class. Consequently, it achieved an excellent clustering map that is very close to the ground truth, with much higher OA, NMI, and Kappa of 96.05%, 88.40% and 0.9466, which further confirms its superiority.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS FOR THE HU IMAGE

Metric	Class	LRR	EDSC	EGCSC	GCOT	GR_RSCNET	HyAE	UNADMM	LGDSC
PA (%)	1	91.77	99.82	**100**	**100**	**100**	98.35	42.96	95.80
	2	9.61	9.80	0	0	**100**	97.78	20.15	**100**
	3	59.40	79.63	99.86	0	**100**	**100**	**100**	**100**
	4	17.65	5.66	0	0.45	0	0	0.23	**98.64**
	5	87.37	84.34	82.89	**100**	99.34	**100**	**100**	**100**
	6	25.98	27.24	0.25	97.10	89.28	98.87	**99.75**	88.90
	7	0	67.92	70.44	70.44	70.44	86.16	64.78	**88.68**
	8	25.45	20.72	73.25	75.53	**99.84**	57.26	0	92.66
	9	60.21	1.31	**62.30**	0	26.44	26.44	0	47.38
	10	3.85	**7.69**	0	0	0	0	0	0
	11	18.46	16.59	95.79	74.07	**100**	**100**	**100**	95.56
	12	19.70	20.30	17.12	36.82	80.61	64.39	0	**100**
UA (%)	1	49.65	43.16	35.98	35.79	89.37	56.69	81.59	**100**
	2	10.25	15.14	0	0	58.93	92.48	**100**	95.89
	3	61.88	59.16	53.45	0	**100**	**100**	50.39	**100**
	4	16.28	7.28	0	4.87	0	0	0	**99.10**
	5	78.57	54.83	62.81	54.83	**100**	**100**	**100**	88.27
	6	41.36	57.29	1.73	47.53	88.83	**92.23**	29.99	84.37
	7	0	77.69	**100**	**100**	**100**	21.71	**100**	**100**
	8	34.89	33.15	75.20	72.91	63.94	**100**	0	81.26
	9	56.65	1.85	98.34	0	95.28	**100**	**100**	**100**
	10	0.42	**0.74**	0	0	0	0	0	0
	11	18.67	18.93	50.74	99.68	98.61	92.44	**100**	**100**
	12	47.97	78.82	99.12	80.73	**100**	**100**	0	**100**
OA (%)		41.53	41.05	52.86	53.14	83.22	78.53	53.31	**92.87**
NMI (%)		44.87	46.58	69.52	69.46	87.45	86.83	69.25	**90.99**
Kappa		0.3558	0.3479	0.4784	0.4745	0.8132	0.7624	0.4720	**0.9206**

TABLE V
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS FOR THE XZ IMAGE

Metric	Class	LRR	EDSC	EGCSC	GCOT	GR_RSCNET	HyAE	UNADMM	LGDSC
PA (%)	1	0	**100**	**100**	**100**	**100**	99.88	**100**	**100**
	2	74.32	77.29	88.91	87.54	87.54	24.52	26.25	**92.43**
	3	21.73	0.71	14.28	73.97	73.97	96.02	**99.18**	89.89
	4	92.73	95.46	81.07	88.58	88.58	**98.57**	90.88	97.81
	5	47.90	95.84	55.60	55.94	55.94	98.91	**99.89**	97.60
UA (%)	1	0	80.47	78.06	89.00	**95.44**	91.60	69.18	89.53
	2	61.51	97.30	66.57	69.81	**100**	**100**	**100**	98.38
	3	18.00	12.72	16.11	63.09	96.23	98.63	**99.79**	95.86
	4	89.00	83.30	82.05	89.47	99.36	99.65	**99.83**	99.76
	5	74.60	79.12	84.72	81.87	57.33	58.15	59.21	**91.16**
OA (%)		64.73	84.02	72.60	79.99	81.08	81.57	79.64	**96.05**
NMI (%)		52.47	77.88	61.19	73.12	79.59	80.38	78.11	**88.40**
Kappa		0.4908	0.7238	0.6312	0.6310	0.7431	0.7501	0.7271	**0.9466**

C. Comparisons With the Other SOTA Deep Clustering Models

In this section, seven other SOTA deep clustering methods for HSIs were selected for further comparison, to verify the efficacy of LGDSC, i.e., hypergraph convolutional subspace clustering (HGCSC) [61], 3D-ACAE [39], CMSCGC [59], pixel super-pixel CL pseudo-label correction (PSCPC) [64], super-pixel spatially regularized diffusion learning (S^2DL) [65], enhanced multi-view contrastive clustering (EMVCC) [46], and DSGCC [43]. These methods include both DSC and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

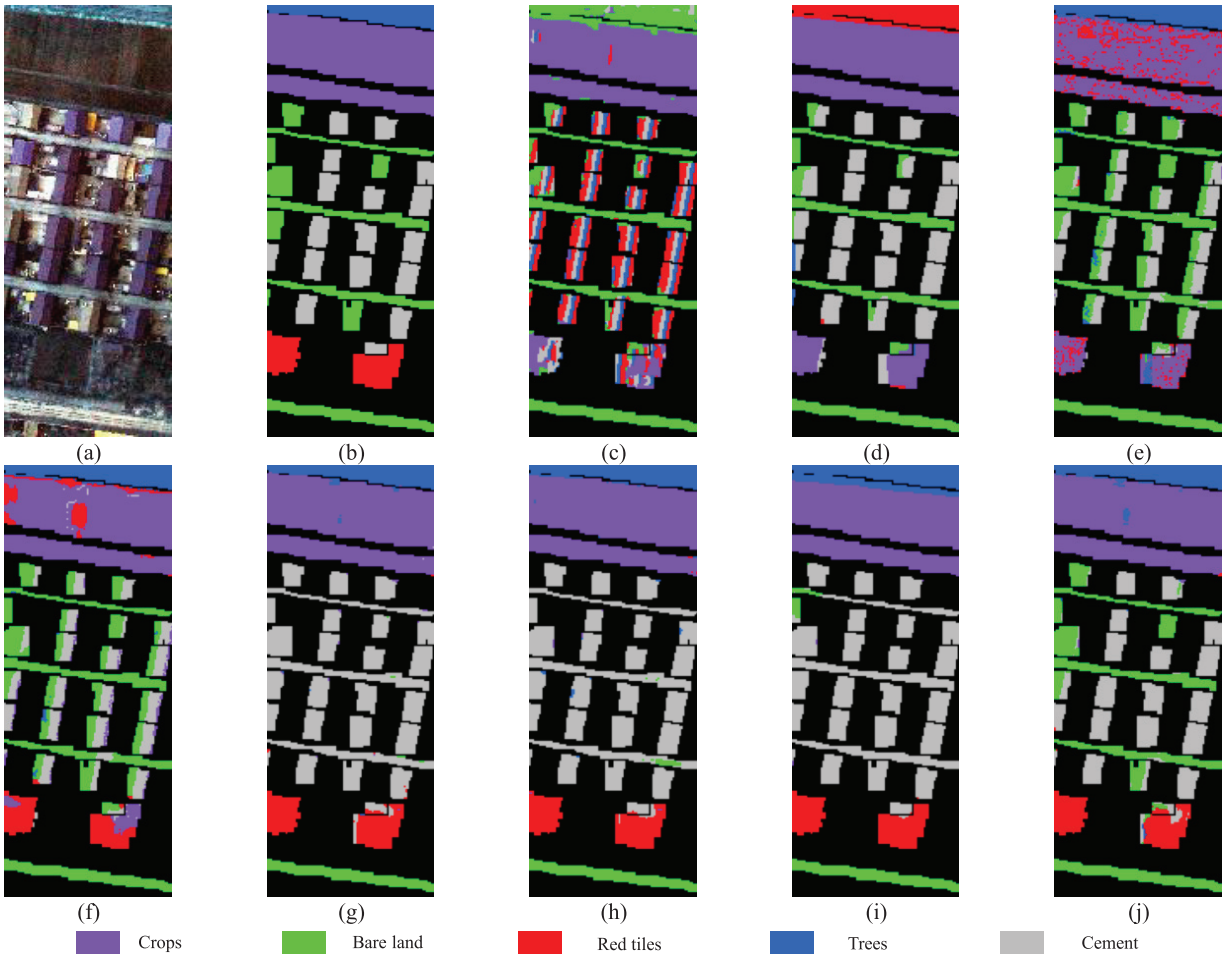


Fig. 7. Clustering results derived by different methods for the XZ image. (a) False color image, (b) ground truth, (c) LRR, (d) EDSC, (e) EGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) LGDSC. (All maps are rotated 90°.)

TABLE VI
COMPARISON WITH THE OTHER SOTA MODELS

Method	IP dataset			PU dataset			HU dataset			XZ dataset		
	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa
HGCSC	\	\	\	92.44	89.21	0.8307	73.15	70.04	0.8309	91.57	88.62	0.8470
3D-ACAE	\	\	\	65.49	65.49	\	\	\	\	\	\	\
CMSCGC	\	\	\	96.69	97.02	0.9565	87.21	**91.68**	0.8580	\	\	\
PSCPC	64.46	64.52	0.6032	\	\	\	\	\	\	\	\	\
S^2DL	64.70	59.10	0.6020	\	\	\	\	\	\	\	\	\
EMVCC	65.13	69.81	0.6184	\	\	\	\	\	\	\	\	\
DSGCC	65.45	61.54	0.6097	\	\	\	\	\	\	\	\	\
LGDSC	**67.41**	**72.90**	**0.6369**	**100**	**100**	**1.0000**	**92.87**	90.99	**0.9206**	**96.05**	**88.40**	**0.9466**

other deep framework-based models. Due to the unavailability of complete source codes, the visual maps cloud not be precisely reproduced. Therefore, following most works [59], [60], [61], [62], the reported quantitative evaluations were employed for direct comparison, as listed in Table VI.

LGDSC presented a clear advantage over the other methods and achieved the best results for all four images, with obvious improvements across most metrics. For the IP image, it improved the OA, NMI, and Kappa by 2.28%, 3.09%, and 1.85%, as compared to the suboptimal EMVCC, with larger gains over the other approaches. For the PU image, the increments were more obvious, with large gains of 3.31%, 2.98%, and 4.35% in OA, NMI, and Kappa over the suboptimal CMSCGC. For the HU and XZ images, LGDSC attained larger OA and Kappa values, with significant gains of 5.66%, 4.48% and 6.26%, 9.96%, compared to the suboptimal methods, while obtaining comparable NMI values. These results further demonstrate that LGDSC is a promising tool for practical applications.

D. Efficiency Comparison of Different Methods

To further explore the practicability of the LGDSC model, the efficiency of different methods was analyzed. Since the baselines include some traditional models, for fair comparison,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
TIME COST OF DIFFERENT METHODS IN CPU (S)

Method	IP	PU	HU	XZ
LRR	16953.84	3548.78	3175.71	12671.41
EDSC	**53.26**	53.57	67.25	254.79
EGCSC	234.63	**52.77**	**60.23**	**159.38**
GCOT	5683.70	5468.07	3201.47	5147.93
GR_RSCNET	4643.08	1954.29	1736.72	3586.34
HyAE	5683.70	5468.07	3201.47	5147.93
UNADMM	14809.08	5842.37	5207.95	20787.88
LGDSC	11934.00	3147.10	5320.84	5261.53

TABLE VIII
TIME COST OF THE DSC METHODS IN GPU (S)

Method	PU	HU
GR_RSCNET	1486.29	**980.48**
HyAE	2018.37	1972.86
UNADMM	**442.86**	1638.35
LGDSC	870.00	2979.00

the time cost of all methods in CPU was first compared on all four datasets, with the time cost of DSC methods in GPU further compared on the PU and HU datasets. Tables VII and VIII give the detailed processing time.

Table VII shows that DSC models generally have much higher time costs than traditional subspace clustering models, except for LRR, due to the complex feature engineering process executed by deep learning. Specifically, EDSC and EGCSC are efficient, while LRR and GCOT are time-consuming. Among the DSC methods, GR_RESNET is relatively efficient, while HyAE has larger time costs on all datasets due to its additional hypergraph structure mining module. Our method is more time-consuming than GR_RSCNET, especially for the IP image, as a result of the introduction of dual CAEs and the SAA module. It has comparative and slightly larger time costs with HyAE on the HU and XZ datasets, respectively, while it is more efficient than HyAE on the PU dataset, due to its fast convergence for this simple scene with regular land object distribution. Comparatively, UNADMM is relatively time-consuming, especially for the first and fourth scenes, because of its complex optimization of the coefficient matrix. Nevertheless, as noticed in Table VIII, all DSC methods are greatly accelerated by GPU, especially for our method and UNADMM, making them more efficient than GR_RESNET on the PU dataset. Overall, LGDSC significantly improves the performance with an acceptable time cost, which can be an attractive tool in real applications.

E. Ablation Studies

Within a typical unsupervised encoder-decoder framework, LGDSC includes two key parts, i.e., dual CAE with joint loss and the SAA module. To verify the effectiveness of each part, systematic ablation experiments were conducted on all four datasets. Both 1-D CAE with single loss and 2-D CAE with single loss were taken as the baselines, with dual CAE with joint loss and the SAA module sequentially added. The detailed results are shown in Table IX.

1) Ablation of Dual CAE With Joint Loss: The dual CAE integrates local-spectral and global-spatial feature learning tasks into a unified framework to fully exploit complementary contents and underlying values contained in HSIs while exploring pixel affinities in multiple ways, thus facilitating feature mining and affinity learning, which determines feature discriminability and affinity completeness, with a significant influence on the clustering performance. From Table IX, it can be seen that by integrating the advantages of two branches, dual CAE with joint loss significantly surpassed 1-D CAE and 2-D CAE for all four images, which clearly illustrates its efficacy. Specifically, it improved the OA by about 7% on the IP dataset, 4% on the PU dataset, 9% on the HU dataset, and 15% on the XZ dataset, over the single CAEs. While for the other two metrics, i.e., NMI and Kappa, the increments were also significant, which indicates the necessity of integrating the complementarity of spectral–spatial and local–global features. In addition, it can be noticed that the 2-D CAE for global-spatial feature extraction outperformed the 1-D CAE for local-spectral feature extraction, except for the HU image. It is because this scene has many small objects, so the small window in the local-spectral branch guarantees the purity of ground objects while the large window in the global-spatial branch introduces certain unrelated interference and noise, resulting in downgraded performance. This further highlights the strong potential of spectral information in HSIs.

2) Ablation of the SAA Module: The SAA module can precisely fuse complementary pixel affinities from differentiated coefficient matrices learned by the local-spectral and global-spatial branches in a selective way, rather than simply fusing them by element-wise addition as in dual CAE with joint loss. This results in a more complete and clustering-friendly coefficient matrix. Hence, it determines the accuracy of pixel affinities and strongly affects the clustering performance. As noticed, precisions were further improved to some degree, assisted by the SAA module. It improved the OA by 0.62% on the IP dataset, 0.33% on the PU, 5.32% on the HU, and 0.74% on the XZ dataset, with clear gains also observed for NMI and Kappa, which confirms the efficacy of the SAA module.

F. Parameter Analysis

In LGDSC, two key parameters, i.e., window size and kernel size, play an important role in feature extraction. In addition, three regularization parameters in the joint loss determine the contribution of each term. Hence, a series of experiments was performed to indicate their specific influence on the clustering accuracy, with all four datasets used. Figs. 8–12 illustrate the accuracy change trends with variations of each parameter.

1) Influence of Window Size: The window size determines the amount of information and the feature learning range. As the window size increases, data richness rises, but the risk of interference and noise increases at the same time. Therefore, it has a certain impact on feature quality and subsequently affects clustering accuracy. Fig. 8 illustrates the influence of both

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION STUDIES OF LGDSC FOR ALL FOUR IMAGES

Network component	IP dataset			PU dataset			HU dataset			XZ dataset		
	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa
1D CAE	57.71	59.75	0.5178	88.13	89.81	0.8462	78.67	88.91	0.7622	75.07	68.94	0.6635
2D CAE	59.74	70.8	0.5479	95.87	94.15	0.9454	72.17	83.51	0.6886	80.50	78.89	0.7366
Dual CAE	66.79	70.91	0.6238	99.67	98.99	0.9957	87.55	88.15	0.8622	95.31	87.72	0.9369
Dual CAE+SAA	**67.41**	**72.90**	**0.6369**	**100**	**100**	**1.00**	**92.87**	**90.99**	**0.9206**	**96.05**	**88.40**	**0.9466**

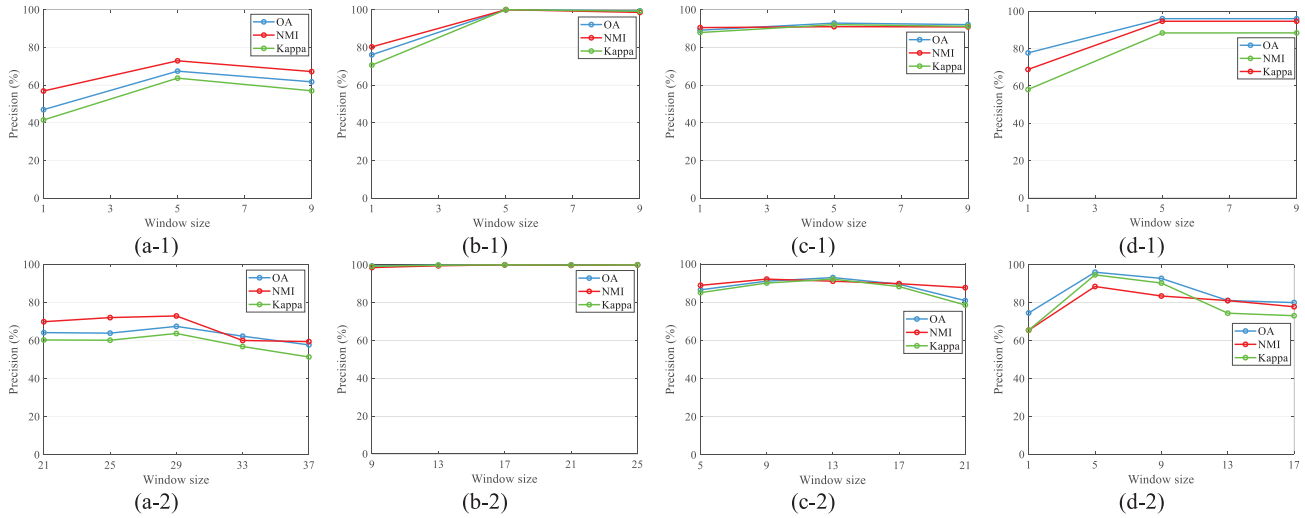


Fig. 8. Impact of window size on clustering accuracy. (a)–(d) IP, PU, HU, and XZ images, with (1) and (2) representing 1-D CAE and 2-D CAE, respectively.

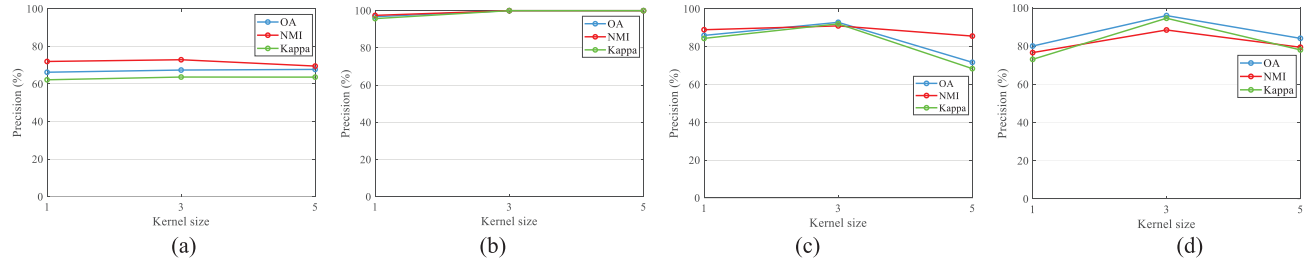


Fig. 9. Impact of kernel size on clustering accuracy for all four images. (a) IP image, (b) PU image, (c) HU image, and (d) XZ image.

local-spectral and global-spatial window sizes on clustering accuracy for all four images.

Fig. 8 shows that window size exerts a large influence on clustering accuracy. For the local-spectral window size, a consistent trend is observed across all four images. When the window size increases, the accuracy indicated by OA, NMI, and Kappa first rises and then declines, with the optimal value fixed at 5×5. A similar change trend can be found for the global-spatial window size, but the optimal value varied across different images. A larger window size of 29×29 is more appropriate for the IP image with smooth ground objects, while small window sizes of 13×13 and 5×5 are more suitable for the HU and XZ images with small objects and complex distributions, to ensure the purity of features. For the PU image, which has composite characteristics, a moderate value of 17×17 performs best. Overall, LGDSC presented stable performance, with accuracy fluctuating within a limited range. The optimal value exhibits a consistent pattern that may be applicable in other scenarios.

2) Influence of Kernel Size: The convolutional kernel size defines the range of the receptive field and determines the strength of neighborhood correlations. An increase in kernel size allows the capture of more distant pixel correlations. Hence, kernel size has a large impact on feature expressivity. Herein, we unified the kernel size for 1-D and 2-D convolutions. Fig. 9 depicts its influence on the accuracy for all four images.

Similarly, the kernel size has a definite impact on the clustering accuracy, but with a unified change trend for all four images. The accuracy first rises and then falls when kernel size increases, with the optimal value fixed at 3. The reason may be that a small kernel size captures limited neighborhood contexts and reduces feature expressivity, whereas a large kernel size may introduce irrelated interference from different classes. This finding gives clear guidance for real applications.

3) Influence of Regularization Parameters: In the joint loss function, three regularization parameters, i.e., α, β, and γ, are used to balance different constraints. Specifically, α is used

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

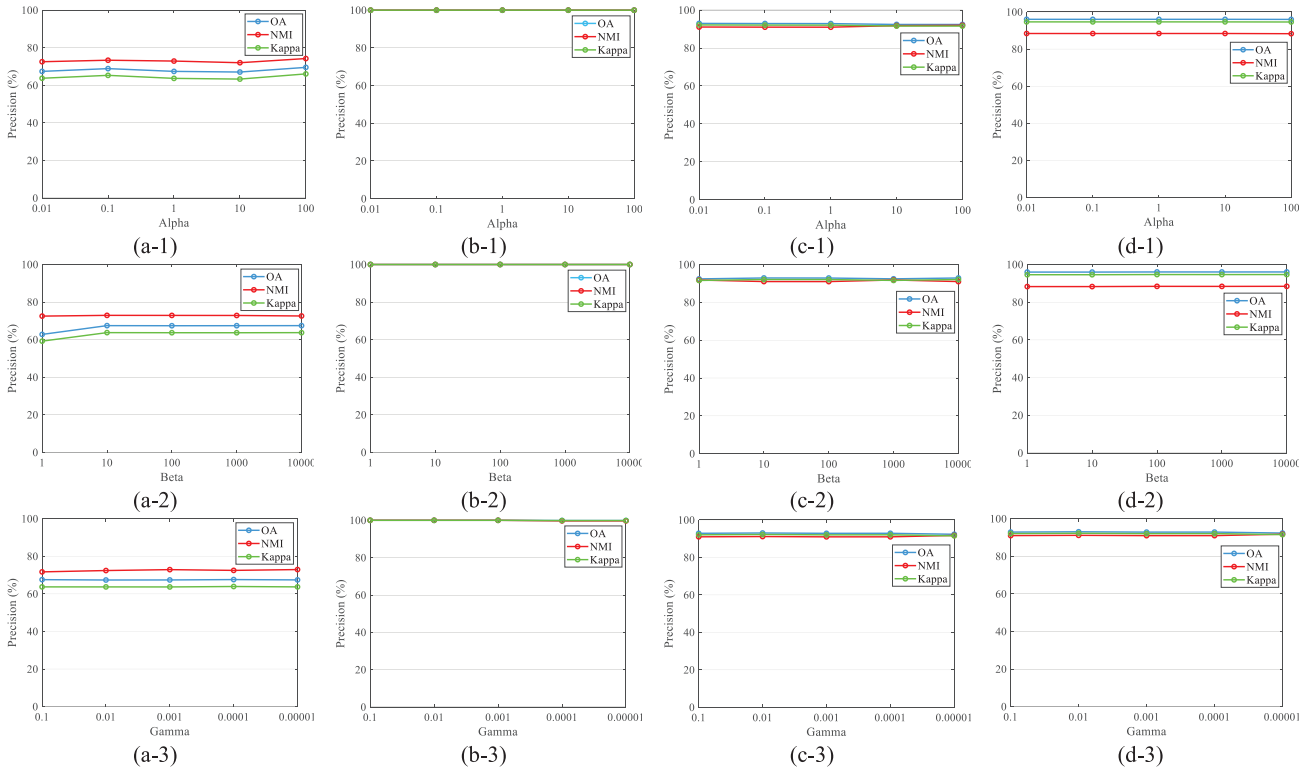


Fig. 10. Impact of regularization parameters on clustering accuracy. (a)–(d) IP, PU, HU, and XZ images, with (1)–(3) representing the impact of alpha, beta, and gamma, respectively.

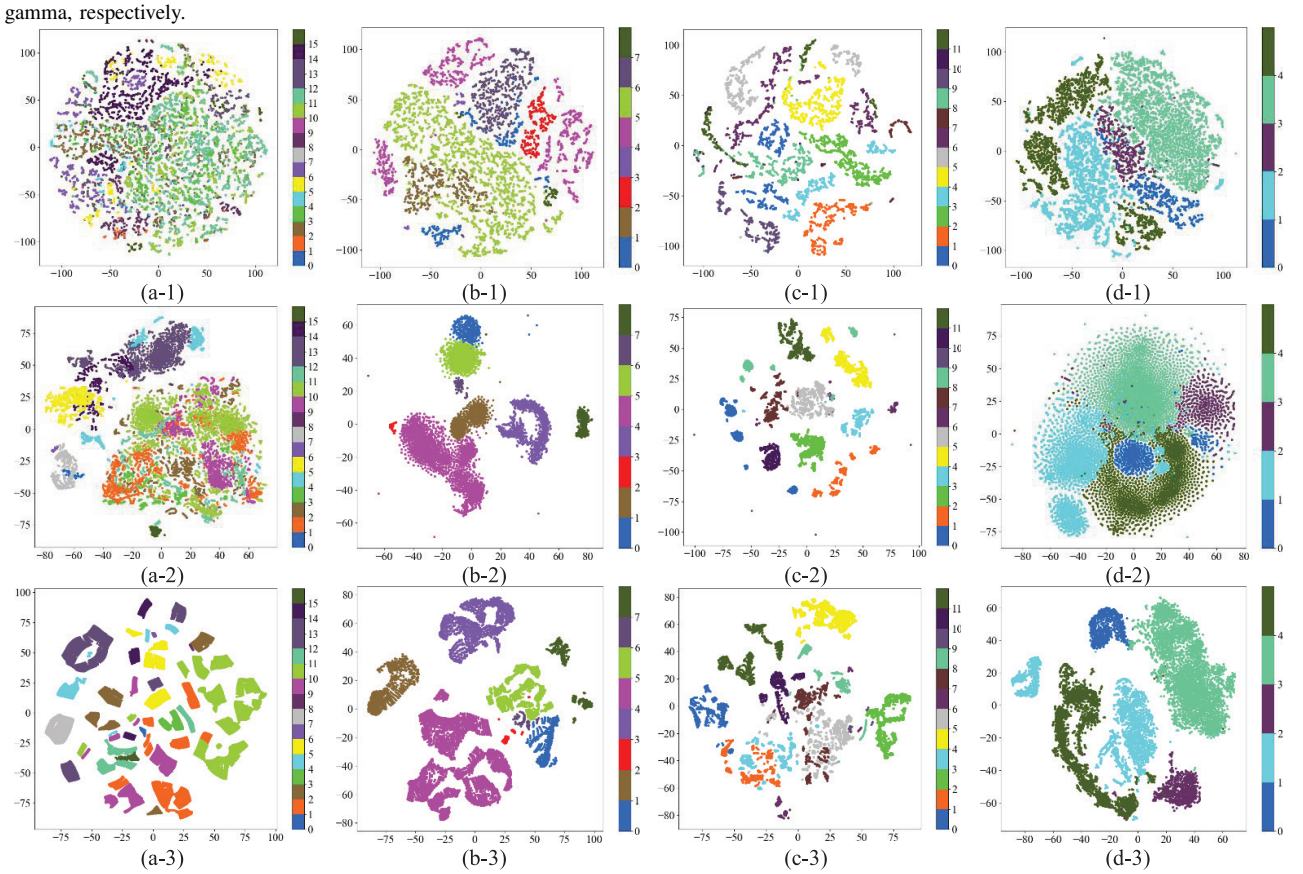


Fig. 11. Feature space visualization based on t-SNE. (a)–(d) IP, PU, HU, and XZ images, with (1)–(3) representing the original, deepest local-spectral and deepest global-spatial feature spaces, respectively.

to trade off data reconstruction error and gets values from the range of [0.01, 0.1, 1, 10, 100]. β is used to control self-expression error and takes a value from the range of [1, 10, 100, 1000, 10000]. γ is used to adjust the strength of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

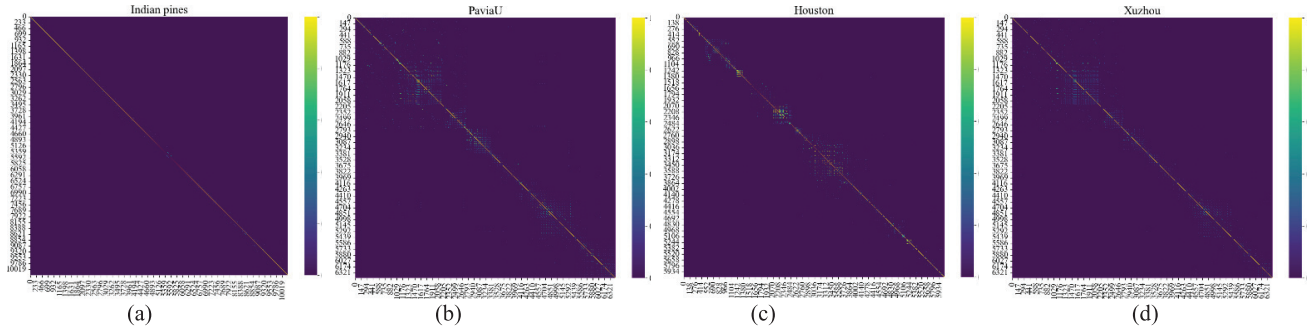


Fig. 12. Affinity matrix visualization learned by LGDSC for all four images. (a) IP dataset, (b) PU dataset, (c) HU dataset, and (d) XZ dataset.

structural prior for the coefficient matrix and selects values from the range of [0.1, 0.01, 0.001, 0.0001, 0.00001]. Fig. 10 shows their impact on the final clustering accuracy.

From Fig. 10, it can be clearly observed that LGDSC is relatively robust to these hyperparameters, maintaining stable performance across a wide range. The clustering accuracy has little fluctuation. By comprehensively considering all the results, a unified set of hyperparameters can be obtained, i.e., $\alpha = 1, \beta = 100$ and $\gamma = 0.001$, which provides clear guidance for practical applications and further increases the practicability of LGDSC.

G. Visualization of Feature Space and Adjacency Matrix

To further verify the efficacy of LGDSC in feature mining and affinity capturing, we visualized the deepest encoding features derived from the dual CAE and the adjacency matrix generated by the SAA module, as shown in Figs. 11 and 12.

Specifically, the encoding features form the foundation of self-expression, and their discriminability and expressivity directly determine the precision and quality of self-expression learning. To intuitively analyze the performance of LGDSC in feature extraction, the deepest local-spectral and global-spatial features learned by 1-D CAE and 2-D CAE were visualized using t-distributed stochastic neighbor embedding (t-SNE), with the original feature space of HSIs employed for comparison [59], [60], [61], [62].

As illustrated in Fig. 11, feature engineering greatly changes the sample distribution in the feature space formed by the first two principal components. In the original feature space, samples from different classes are mixed together without clear boundaries, while samples from the same class are scattered and separated into several parts, creating large difficulties for clustering. By comparison, after deeply processing, samples in both the local-spectral and global-spatial feature space exhibit a more regular and compact distribution, with distinct clusters and clear boundaries, which effectively reveals the intrinsic structure of HSIs and facilitates more precise affinity mining. Furthermore, it can be noticed that global-spatial features generally provide a smoother and more separable sample distribution than local-spectral features, owing to their richer contextual information and higher discriminability, as observed in the IP, PU, and XZ images. However, the opposite phenomenon is observed for the HU image. The reason could be that this scene contains many small objects with scattered distributions, which reduces the purity of global-spatial features for individual objects and consequently degrades discriminability. This effect is consistent with the results of ablation studies. Generally speaking, LGDSC has a powerful feature extraction capability.

The adjacency matrix captures the underlying relationships between pixels and is the basis of spectral clustering, with a larger value referring to a higher affinity. It is induced by the coefficient matrix with a nonnegative symmetrical structure, whose quality directly affects clustering accuracy. From Fig. 12, it can be clearly observed that adjacency matrices for all four images exhibit a distinct block-diagonal structure with sparse distribution. Herein, each block corresponds to a cluster, with nonzero elements concentrated along the diagonal line and larger values aggregated in the surrounding region. This structure reveals the latent partition pattern, ensuring that pixels are separated into the correct clusters, which demonstrates that LGDSC effectively captures precise pixel affinities.

V. Conclusion

Faced with the shortcomings of current DSC models, this article proposed a novel LGDSC network for HSIs, which fully exploits rich spectral and spatial diagnostic information, together with complementary local and global contents to promote more accurate clustering in complex scenarios. On the one hand, an effective dual CAE with a parallel and symmetrical structure was constructed to focus on local-spectral and global-spatial feature expression learning through 1-D CAE and 2-D CAE, respectively, with a shared self-expression module embedded for unified pixel affinity learning. Herein, a valid joint loss function with biliteral constraints was developed to simultaneously guide local-spectral and global-spatial feature mining. On the other hand, a novel SAA module was designed based on the underlying mechanism of self-expression to precisely fuse favorable affinities learned by differentiated features while eliminating interference to generate a more clustering-friendly adjacency matrix. Experimental results indicate that LGDSC significantly outperformed the other SOTA methods, and could provide a practical solution for real-world applications.

However, there is still some room for improvement. First, by incorporating well-designed transformer modules to directly model global information, rather than relying on an aggregation scheme, more expressive global features can be generated. Second, by developing a more precise self-expression scheme

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

to capture more complete pixel affinities, the clustering accuracy would be further improved. Third, by combining the model with CL to enhance its adaptability to large-scale scenes, it will be more attractive in real applications. These directions will be pursued in our future work.

REFERENCES

[1] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[2] W.-T. Zhang, Y. Bai, S.-D. Zheng, J. Cui, and Z.-Z. Huang, "Tensor transformer for hyperspectral image classification," *Pattern Recognit.*, vol. 163, Jul. 2025, Art. no. 111470.

[3] H. Zhai and J. Zhao, "Two-stream spectral–spatial convolutional capsule network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103614.

[4] H. Zhai and Y. Liu, "Hyperspectral image classification based on Atrous convolution channel attention aided dense convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 5504305.

[5] G. Wang, X. Zhang, Z. Peng, T. Zhang, and L. Jiao, "S^2mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511413.

[6] S. Lorenz, P. Ghamisi, M. Kirsch, R. Jackisch, B. Rasti, and R. Gloaguen, "Feature extraction for hyperspectral mineral domain mapping: A test of conventional and innovative methods," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112129.

[7] Y. Pan, M. Wu, L. Zhang, and J. Chen, "Time series clustering-enabled geological condition perception in tunnel boring machine excavation," *Autom. Construction*, vol. 153, Sep. 2023, Art. no. 104954.

[8] P. Souto-Ceccon, G. Simarro, P. Ciavola, A. Taramelli, and C. Armaroli, "Shoreline detection from PRISMA hyperspectral remotely-sensed images," *Remote Sens.*, vol. 15, no. 8, p. 2117, Apr. 2023.

[9] H. Zhai, H. Zhang, P. Li, and L. Zhang, "Hyperspectral image clustering: Current achievements and future lines," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 35–67, Dec. 2021.

[10] S. Lloyd, "Least squares quantization in PCM," *IEEE Trans. Inf. Theory*, vol. IT-28, no. 2, pp. 129–137, Mar. 1982.

[11] J. C. Bezdek, *Pattern Recognition With Fuzzy Objective Function Algorithms*. Cham, Switzerland: Springer, 2013.

[12] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, Jun. 2014.

[13] J. M. Murphy and M. Maggioni, "Spectral–spatial diffusion geometry for hyperspectral image clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 7, pp. 1243–1247, Jul. 2020.

[14] N. Acito, G. Corsini, and M. Diani, "An unsupervised algorithm for hyperspectral image segmentation based on the Gaussian mixture model," in *Proc. IGARSS. IEEE Int. Geosci. Remote Sens. Symposium*, vol. 6, pp. 3745–3747.

[15] Y. Jiao, Y. Ma, and Y. Gu, "Hyperspectral image clustering based on variational expectation maximization," in *Proc. IEEE 11th Sensor Array Multichannel Signal Process. Workshop (SAM)*, Jun. 2020, pp. 1–5.

[16] Y. Zhong, L. Zhang, and W. Gong, "Unsupervised remote sensing image classification using an artificial immune network," *Int. J. Remote Sens.*, vol. 32, no. 19, pp. 5461–5483, Oct. 2011.

[17] H. Jiao, Y. Zhong, and L. Zhang, "An unsupervised spectral matching classifier based on artificial DNA computing for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4524–4538, Aug. 2014.

[18] A. Hassanzadeh, A. Kaarna, and T. Kauranne, "Sequential spectral clustering of hyperspectral remote sensing image over bipartite graph," *Appl. Soft Comput.*, vol. 73, pp. 727–734, Dec. 2018.

[19] R. Wang, F. Nie, Z. Wang, F. He, and X. Li, "Scalable graph-based clustering with nonnegative relaxation for large hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7352–7364, Oct. 2019.

[20] Z. Cao, H. Xin, R. Wang, and F. Nie, "Superpixel-based bipartite graph clustering enriched with spatial information for hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5505115.

[21] H. Zhai, H. Zhang, L. Zhang, P. Li, and X. Xu, "Spectral–spatial clustering of hyperspectral remote sensing image with sparse subspace clustering model," in *Proc. 7th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[22] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[23] Y. Zhang, X. Wang, X. Jiang, L. Zhang, and B. Du, "Elastic graph fusion subspace clustering for large hyperspectral image," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 7, pp. 6300–6312, Jul. 2025, doi: 10.1109/TCSVT.2025.3538926.

[24] H. Zhai, H. Zhang, L. Zhang, P. Li, and A. Plaza, "A new sparse subspace clustering algorithm for hyperspectral remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 43–47, Jan. 2017.

[25] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Laplacian-regularized low-rank subspace clustering for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1723–1740, Mar. 2019.

[26] J. Xu, J. E. Fowler, and L. Xiao, "Hypergraph-regularized low-rank subspace clustering using superpixels for unsupervised spatial–spectral hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 871–875, May 2021.

[27] J. Chen, S. Liu, Z. Zhang, and H. Wang, "Diffusion subspace clustering for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6517–6530, 2023.

[28] H. Zhai, H. Zhang, X. Xu, L. Zhang, and P. Li, "Kernel sparse subspace clustering with a spatial max pooling operation for hyperspectral remote sensing data interpretation," *Remote Sens.*, vol. 9, no. 4, p. 335, Apr. 2017.

[29] H. Zhao, F. Zhou, L. Bruzzone, R. Guan, and C. Yang, "Superpixel-level global and local similarity graph-based clustering for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[30] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Total variation regularized collaborative representation clustering with a locally adaptive dictionary for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 166–180, Jan. 2019.

[31] Z. Chen, C. Zhang, T. Mu, and Y. He, "Tensorial multiview subspace clustering for polarimetric hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528513.

[32] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[33] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Sparsity-based clustering for large hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10410–10424, Dec. 2021.

[34] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Nonlocal means regularized sketched reweighted sparse and low-rank subspace clustering for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4164–4178, May 2021.

[35] Z. Zhang, Y. Cai, W. Gong, X. Liu, C. Zeng, and G. Yu, "MMAGL: Multiobjective multiview attributed graph learning for joint clustering of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5505314.

[36] Y. Yao, B. Peng, T. Qin, Y. Gu, N. Ling, and J. Lei, "Hypergraph contrastive learning for large-scale hyperspectral image clustering," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 7, pp. 7090–7100, Jan. 2025.

[37] X. Li and J. Liu, "Graph convolutional and random Fourier feature mapping for hyperspectral image clustering," *J. Supercomput.*, vol. 81, no. 1, p. 193, Jan. 2025.

[38] M. Zeng, Y. Cai, X. Liu, Z. Cai, and X. Li, "Spectral–spatial clustering of hyperspectral image based on Laplacian regularized deep subspace clustering," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2694–2697.

[39] Z. Zheng, S. Zhang, H. Song, and Q. Yan, "Deep clustering using 3D attention convolutional autoencoder for hyperspectral image analysis," *Sci. Rep.*, vol. 14, no. 1, p. 4209, Feb. 2024.

[40] S. Liu and H. Wang, "Deep subspace clustering preserving distribution for hyperspectral images," *Electron. Lett.*, vol. 61, no. 1, p. 70262, Jan. 2025.

[41] Y. Cai, M. Zeng, Z. Cai, X. Liu, and Z. Zhang, "Graph regularized residual subspace clustering network for hyperspectral image clustering," *Inf. Sci.*, vol. 578, pp. 85–101, Nov. 2021.

[42] Y. Cai, Z. Zhang, Z. Cai, X. Liu, and X. Jiang, "Hypergraph-structured autoencoder for unsupervised and semisupervised classification of hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[43] J. Chen, S. Liu, and H. Wang, "Dual smooth graph convolutional clustering for large-scale hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6825–6840, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.

[44] X. Hu, T. Li, T. Zhou, and Y. Peng, "Deep spatial–spectral subspace clustering for hyperspectral images based on contrastive learning," *Remote Sens.*, vol. 13, no. 21, p. 4418, Nov. 2021.
[45] Y. Cai et al., "Superpixel contracted neighborhood contrastive subspace clustering network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530113.
[46] F. Luo, Y. Liu, X. Gong, Z. Nan, and T. Guo, "EMVCC: Enhanced multi-view contrastive clustering for hyperspectral images," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 6288–6296.
[47] F. Luo, Y. Liu, Y. Duan, T. Guo, L. Zhang, and B. Du, "SDST: Self-supervised double-structure transformer for hyperspectral images clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511614.
[48] Z. Zhang, H. Wang, S. Liu, J. Chen, Z. Zhang, and S. Wang, "Convolution-augmented transformer network for hyperspectral image subspace clustering," *Earth Sci. Informat.*, vol. 16, no. 3, pp. 2439–2453, Sep. 2023.
[49] Y. Lin, H. Liu, X. Yu, and C. Zhang, "Leveraging transformer-based autoencoders for low-rank multi-view subspace clustering," *Pattern Recognit.*, vol. 161, May 2025, Art. no. 111331.
[50] X. Li, N. Nadisic, S. Huang, and A. Pižurica, "Unfolding ADMM for enhanced subspace clustering of hyperspectral images," in *Proc. 32nd Eur. Signal Process. Conf. (EUSIPCO)*, Aug. 2024, pp. 2022–2026.
[51] A. Li, A. Qin, Z. Shang, and Y. Y. Tang, "Spectral–spatial sparse subspace clustering based on three-dimensional edge-preserving filtering for hyperspectral image," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 33, no. 3, Mar. 2019, Art. no. 1955003.
[52] Q. Yan, Y. Ding, Y. Xia, Y. Chong, and C. Zheng, "Class probability propagation of supervised information based on sparse subspace clustering for hyperspectral images," *Remote Sens.*, vol. 9, no. 10, p. 1017, Sep. 2017.
[53] L. Tian, Q. Du, I. Kopriva, and N. Younan, "Spatial–spectral based multi-view low-rank sparse sbuspace clustering for hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 8488–8491.
[54] F. de Morsier, M. Borgeaud, V. Gass, J.-P. Thiran, and D. Tuia, "Kernel low-rank and sparse graph for unsupervised and semi-supervised classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3410–3420, Jun. 2016.
[55] S. Huang, H. Zhang, and A. Pižurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.
[56] Y. Ding et al., "Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536016.
[57] S. Liu and H. Wang, "Graph convolutional optimal transport for hyperspectral image spectral clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4414013.
[58] T. Han, S. Niu, X. Gao, W. Yu, N. Cui, and J. Dong, "Deep low-rank graph convolutional subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533513.
[59] R. Guan et al., "Contrastive multiview subspace clustering of hyperspectral images based on graph convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510514.
[60] C. Zhang, Y. Zhao, and J. Wang, "Transformer-based dynamic fusion clustering network," *Knowl.-Based Syst.*, vol. 258, Dec. 2022, Art. no. 109984.
[61] Z. Zhang, Y. Cai, W. Gong, P. Ghamisi, X. Liu, and R. Gloaguen, "Hypergraph convolutional subspace clustering with multihop aggregation for hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 676–686, 2022.
[62] G. Liu, Z. Lin, and Y. Yu, "Robust subspace segmentation by low-rank representation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 663–670.
[63] P. Ji, M. Salzmann, and H. Li, "Efficient dense subspace clustering," in *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, Mar. 2014, pp. 461–468.
[64] R. Guan, Z. Li, X. Li, and C. Tang, "Pixel-superpixel contrastive learning and pseudo-label correction for hyperspectral image clustering," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 6795–6799.
[65] K. Cui et al., "Superpixel-based and spatially regularized diffusion learning for unsupervised hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4405818.



Han Zhai (Member, IEEE) received the B.S. degree in remote sensing science and technology from Shandong University of Science and Technology, Qingdao, China, in 2014, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2019.

He has been an Associate Professor with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, since 2019. He has published a series of high-quality articles and patents. His research interests include remote sensing intelligent recognition, hyperspectral information processing, artificial intelligence, and remote sensing applications.

Dr. Zhai has been a reviewer of more than ten international journals, such as *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON NEURAL NETWORK AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and so on.



Jun Yu (Student Member, IEEE) received the B.S. degree in geographic information science from Central South University, Changsha, China, in 2023. He is currently pursuing the M.S. degree with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China.

His research interests include deep learning and hyperspectral image processing, and clustering.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:18:52 UTC from IEEE Xplore. Restrictions apply.